

SHALLOW MINORS, GRAPH PRODUCTS, AND BEYOND-PLANAR GRAPHS*

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Abstract. The planar graph product structure theorem of Dujmović et al. [*J. ACM*, 67 (2020), 22] states that every planar graph is a subgraph of the strong product of a graph with bounded treewidth and a path. This result has been the key tool to resolve important open problems regarding queue layouts, nonrepetitive colorings, centered colorings, and adjacency labeling schemes. In this paper, we extend this line of research by utilizing shallow minors to prove analogous product structure theorems for several beyond-planar graph classes. The key observation that drives our work is that many beyond-planar graphs can be described as a shallow minor of the strong product of a planar graph with a small complete graph. In particular, we show that powers of bounded degree planar graphs, k -planar, (k, p) -cluster planar, fan-planar, and k -fan-bundle planar graphs have such a shallow-minor structure. Using a combination of old and new results, we deduce that these classes have bounded queue-number, bounded nonrepetitive chromatic number, polynomial p -centered chromatic numbers, linear strong coloring numbers, and cubic weak coloring numbers. In addition, we show that k -gap planar graphs have at least exponential local treewidth and, as a consequence, cannot be described as a subgraph of the strong product of a graph with bounded treewidth and a path.

Key words. product structure, shallow minors, beyond planar

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1. Introduction. Many results in structural graph theory describe complex graph classes in terms of simpler classes. For example, Robertson and Seymour’s graph minor structure theorem [77] states that every graph in a proper minor-closed class can be described by a tree-decomposition of graphs that are almost embeddable on a surface with bounded Euler genus. Recently, graph products have emerged as a powerful tool in describing graphs in a complex class. Say a graph H is *contained* in a graph G if H is isomorphic to a subgraph of G , written $H \subseteq G$. Dujmović et al. [28] established that every planar graph is contained in the strong product of a graph with bounded treewidth and a path.

THEOREM 1 (see [28]). *Every planar graph is contained in $H \boxtimes P$ for some planar graph H of treewidth at most 8 and for some path P .*

This breakthrough has been the key tool to resolve several major open problems regarding queue layouts [28], nonrepetitive colorings [27], centered colorings [20], adjacency labeling schemes [11, 26, 41], twin-width [13], infinite graphs [55], and comparable box dimension [33]. Treewidth measures how similar a graph is to a tree and is an important parameter in algorithmic and structural graph theory (see [8, 48, 75]). Graphs with bounded treewidth are a simpler class of graphs compared

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to planar graphs. Theorem 1 therefore reduces problems on a complicated class of graphs (planar graphs) to a simpler class of graphs (bounded treewidth).

Motivated by Theorem 1, Bose et al. [14] defined the *row treewidth*, $\text{rtw}(G)$, of a graph G to be the minimum treewidth of a graph H such that $G \subseteq H \boxtimes P$ for some path P . Theorem 1 says that planar graphs have row treewidth at most 8. Ueckerdt, Wood, and Yi [80] strengthened Theorem 1 by improving the upper bound to 6. Other graph classes that have bounded row treewidth include graphs with bounded Euler genus [24, 28], apex-minor-free graphs [28], (g, d) -map graphs [31], (g, δ) -string graphs [31], (g, k) -planar graphs [31], squaregraphs [52], and powers of bounded degree planar graphs [31].

In this paper, we extend this line of research by utilizing shallow minors and graph products to demonstrate that several beyond-planar graph classes have bounded row treewidth. The study of beyond-planar graphs is a vibrant research topic within the graph drawing community (see the surveys [22, 54]). A shallow minor of a graph is obtained by contracting subgraphs with small radii then deleting vertices and edges. While beyond-planar graph classes often contain arbitrarily large complete graphs as minors, they are “sparse” under shallow minors. The key observation that drives our work is that many beyond-planar graphs can be described as a shallow minor of the strong product of a planar graph with a small complete graph. In particular, we show that the following beyond-planar classes have such a shallow-minor structure: powers of bounded degree planar graphs (subsection 4.1); (g, k) -planar graphs (subsection 4.2); (g, δ) -string graphs (subsection 4.3); fan-planar graphs (subsection 4.4); and k -fan-bundle-planar graphs (subsection 4.5). To prove these results, we first planarize the given graph. For most of the above graph classes, we use the standard planarization technique of inserting dummy vertices at crossing points. For fan-planar graphs, however, our planarization procedure is more substantial and is of independent interest (see Lemma 33).

A key contribution of this paper is to show that row treewidth is “well-behaved” under shallow minors (see subsection 3.2). In particular, we show that if G has bounded row treewidth and H is an r -shallow minor of $G \boxtimes K_\ell$, then H has bounded row treewidth (see Theorem 10). We conclude that the above beyond-planar graph classes have bounded row treewidth. For fan-planar graphs and k -fan-bundle-planar graphs, it was not known whether they had a product structure. For the other graph classes, Dujmović et al. [31] used the concept of shortcut systems to show that they have bounded row treewidth. However, shortcut systems are limited in that they only apply to graph classes with linear crossing number (see Lemma 5). Shallow minors subsume and generalize shortcut systems. In particular, the results for fan-planar graphs use shallow minors in their full generality since fan-planar graphs have superlinear crossing numbers.

Having established that the above classes have a product structure, we employ tools from graph product structure theory to deduce important structural properties of these classes. In particular, we show that each of these classes has bounded queue-number (subsection 3.3), bounded nonrepetitive chromatic number (subsection 3.4), polynomial p -centered chromatic numbers (subsection 3.5), linear strong coloring numbers (subsection 3.7), and cubic weak coloring numbers (subsection 3.7).

Motivated by the application of shallow minors in describing graph classes, we further explore the structural properties of shallow minors. Since shallow minors are fundamental to graph sparsity theory (see [67]), many papers have considered their properties including the grad function [47, 67], separators [36], twin-width [12], and layered treewidth [30]. We continue this line of research by demonstrating that the queue-number, strong-coloring numbers, and weak coloring numbers are

“well-behaved” under shallow minors. For some of these parameters, we are then able to obtain stronger bounds for the aforementioned beyond-planar graph classes using their shallow-minor structure directly instead of going via product structure. Our result for coloring numbers, which is of independent interest, states that if H is an r -shallow minor of $G \boxtimes K_\ell$, then

$$\text{scol}_s(H) \leq \ell \text{scol}_{2rs+2r+s}(G) \quad \text{and} \quad \text{wcol}_s(H) \leq \ell \text{wcol}_{2rs+2r+s}(G).$$

Our final results show limitations of the above methods. In particular, we show that k -gap planar, fan-crossing free, 3-quasi planar, and right angle crossing (RAC) graphs cannot be described as a shallow minor of the strong product of a planar graph with a small complete graph. Indeed, we show that 1-gap planar graphs have at least exponential local treewidth and thus have unbounded row treewidth (see subsection 5.1). For fan-crossing free, 3-quasi planar, and RAC graphs, we draw on known results to conclude that these classes are somewhere dense (see subsection 5.2). As such, they have unbounded row treewidth, unbounded stack-number, and unbounded queue-number.

2. Preliminaries. We now introduce the relevant notation and background for this paper. See the textbook by Diestel [23] for further background.

2.1. Notation. Let G be a finite, simple, and undirected graph. For $v \in V(G)$, let $N_G(v) := \{u \in V(G) : uv \in E(G)\}$ and $N_G[v] := N_G(v) \cup \{v\}$. When the graph G is clear, we drop the subscript G and use the notation $N(v)$ and $N[v]$. Let $\Delta(G)$ and $\delta(G)$ respectively denote the maximum and the minimum degree of G . A *vertex c -coloring* of G is any function $\phi : V(G) \rightarrow C$ where $|C| \leq c$, and an *edge c -coloring* of G is any function $\phi : E(G) \rightarrow C$ where $|C| \leq c$. For an integer $k \geq 1$, the k th power of G , G^k , is the graph with $V(G^k) := V(G)$ and $uv \in E(G^k)$ if $\text{dist}_G(u, v) \leq k$ and $u \neq v$. The *complement* of G is denoted by $\overline{G}$. Thus $\overline{K_n}$ is the edgeless graph on n vertices. A *class* of graphs $\mathcal{G}$ is a set of graphs that is closed under isomorphism. $\mathcal{G}$ is *hereditary* if it is closed under induced subgraphs and *minor-closed* if it is closed under minors. For integers a, b where $a \leq b$, let $[a, b] := \{a, a+1, \dots, b\}$ and $[b] := [1, b]$.

2.2. Treewidth. For a graph G , a *tree-decomposition* of G consists of a collection $\mathcal{B} = \{B_x \subseteq V(G) : x \in V(T)\}$ of *bags* indexed by a tree T such that (i) for each $v \in V(G)$, $T[\{x \in V(T) : v \in B_x\}]$ is a nonempty subtree of T ; and (ii) for each $uv \in E(G)$, there is a node $x \in V(T)$ such that $u, v \in B_x$. We denote a tree-decomposition by the pair $(T, \mathcal{B})$. The *width* of $(T, \mathcal{B})$ is $\max\{|B_x| - 1 : x \in V(T)\}$. The *treewidth* of a graph G , $\text{tw}(G)$, is the minimum width of a tree-decomposition of G . Tree-decompositions were introduced by Robertson and Seymour [76].

2.3. Graph products. Let G_1 and G_2 be graphs. A *graph product* $G_1 \bullet G_2$ is defined with vertex set $V(G_1 \bullet G_2) := \{(a, v) : a \in V(G_1), v \in V(G_2)\}$. The *lexicographic product* $G_1 \circ G_2$ consists of edges of the form $(a, v)(b, u)$ where either $ab \in E(G_1)$, or $a = b$ and $uv \in E(G_2)$. The *strong product* $G_1 \boxtimes G_2$ consists of edges of the form $(a, v)(b, u)$ where either $ab \in E(G_1)$ and $v = u$ or $uv \in E(G_2)$, or $a = b$ and $uv \in E(G_2)$. Observe that $G_1 \circ G_2 \subseteq G_1 \boxtimes G_2$. We frequently make use of the well-known fact that $\text{tw}(G \boxtimes K_n) \leq (\text{tw}(G) + 1)n - 1$ for every graph G and integer $n \geq 1$ (see [10] for an implicit proof). For a graph product $\bullet \in \{\circ, \boxtimes\}$, graph class $\mathcal{G}$, and graph H , let $\mathcal{G} \bullet H$ denote the class of graphs isomorphic to graphs in $\{G \bullet H : G \in \mathcal{G}\}$.

In addition to Theorem 1, Dujmović et al. [28] proved the following product structure theorem for planar graphs.

THEOREM 2 (see [28]). *Every planar graph is contained in $H \boxtimes P \boxtimes K_3$ for some planar graph H of treewidth at most 3 and for some path P .*

The *Euler genus* of the orientable surface with h handles is $2h$. The *Euler genus* of the nonorientable surface with c cross-caps is c . The *Euler genus* of a graph G is the minimum integer $k \geq 0$ such that G embeds in a surface of Euler genus k ; see [66] for more about graph embeddings in surfaces. Strengthening an earlier result of Dujmović et al. [28], Distel et al. [24] proved the following product structure theorem for graphs on surfaces.

THEOREM 3 (see [24, 28]). *Every graph of Euler genus g is contained in $H \boxtimes P \boxtimes K_{\max\{2g, 3\}}$ for some planar graph H of treewidth at most 3 and for some path P .*

2.4. Beyond-planar graphs. Beyond-planar graphs is a vibrant research topic that studies graph classes defined by drawings that forbid certain crossing configurations. See the recent survey by Didimo et al. [22] as well as the monograph by Hong and Tokuyama [54]. A key objective of this paper is to understand the global structure of beyond-planar graphs.

An *embedded graph* is a graph G with $V(G) \subset \mathbb{R}^2$ in which each edge $uv \in E(G)$ is a curve in $\mathbb{R}^2$ with endpoints u and v and not containing any vertex of G in its interior. A *crossing* in G is a triple $(p, \{uv, xy\})$ with $p \in \mathbb{R}^2$, and $uv, xy \in E(G)$ such that $p \in (uv \cap xy) \setminus \{u, v, x, y\}$. G is *plane* if it has no crossings. G is *simple* if any two edges share at most one point in common, including endpoints. The *crossing number*, $\text{cr}(G)$, of a graph G is the minimum number of crossings in an embedded graph isomorphic to G . A graph class $\mathcal{G}$ has *linear crossing number* if there exists a constant $c > 0$ such that $\text{cr}(G) \leq c|V(G)|$ for every $G \in \mathcal{G}$. By the crossing lemma [1, 64], this is equivalent to there being a constant $c' > 0$ such that $\text{cr}(G) \leq c'|E(G)|$ for every $G \in \mathcal{G}$.

We now introduce the beyond-planar graphs that are relevant to this paper. An embedded graph G is:

- *k-planar* if each edge of G is involved in at most k crossings [72];
- *k-quasi planar* if every set of k edges do not mutually cross;
- *k-gap planar* if every crossing can be charged to one of the two edges involved so that at most k crossings are charged to each edge [5];
- *fan-crossing free* if for each edge $e \in E(G)$, the edges that cross e form a matching [18];
- *fan-planar* if for each edge $e \in E(G)$ the edges that cross e have a common end-vertex and they cross e from the same side (when directed away from their common end-vertex) [58]; or
- *RAC* (right angle crossing) if each edge is drawn as a straight line segment and edges cross at right angles [21].

A graph is *k-planar*, *k-quasi planar*, *k-gap planar*, *fan-crossing free*, *fan-planar*, or *RAC* if it is respectively isomorphic to an embedded graph that is *k-planar*, *k-quasi planar*, *k-gap planar*, *fan-crossing free*, *fan-planar*, or *RAC*.

2.5. Shallow minors and graph sparsity. Let G , H , and J be graphs and let $r \geq 0$ be an integer and $s \geq 0$ be a half-integer (that is, $2s$ is an integer). H is a *minor* of G if a graph isomorphic to H can be obtained from G by vertex deletion, edge deletion, and edge contraction. A *model* of H in G is a function μ with domain $V(H)$ such that (i) $\mu(v)$ is a connected subgraph of G ; (ii) $\mu(v) \cap \mu(w) = \emptyset$ for all distinct $v, w \in V(H)$; and (iii) $\mu(v)$ and $\mu(w)$ are adjacent for every edge $vw \in E(H)$. It is folklore that H is a minor of G if and only if G contains a model of H . If there

exists a model μ of H in G such that $\mu(v)$ has radius at most r for all $v \in V(H)$, then H is an r -shallow minor of G . If there is a model μ of H in G such that $|V(\mu(v))| \leq r$ for all $v \in V(H)$, then H is an r -small minor of G . We say that J is an r -subdivision of H if J can be obtained from H by replacing each edge uv of H by a path of length at most $r+1$ with end-vertices u and v . We say that H is an s -shallow topological minor of G if a subgraph of G is isomorphic to a $2s$ -subdivision of H . Note that if H is an s -shallow topological minor of a graph G , then H is an r -shallow minor of G whenever $s \leq r$.

Shallow minors and shallow topological minors are fundamental to graph sparsity theory (see [67]). In particular, they define bounded expansion, nowhere dense, and somewhere dense graph classes. For an integer $r \geq 0$ and graph G , let $\nabla_r(G)$ be the greatest average degree of an r -shallow minor of G . A hereditary graph class $\mathcal{G}$ has *bounded expansion* with *expansion function* $f_{\mathcal{G}} : \mathbb{N} \cup \{0\} \rightarrow \mathbb{R}$ if $\nabla_r(G) \leq f_{\mathcal{G}}(r)$ for every $r \geq 0$ and graph $G \in \mathcal{G}$. If $f_{\mathcal{G}}$ is polynomial, then $\mathcal{G}$ has *polynomial expansion*. A hereditary graph class $\mathcal{G}$ is *somewhere dense* if there exists an integer $r \geq 0$ such that every graph H is an r -shallow minor of some graph $G \in \mathcal{G}$. If $\mathcal{G}$ is not somewhere dense, then it is *nowhere dense*.

Bounded expansion is a robust measure of sparsity with many characterizations [67, 68, 86]. For example, a graph class $\mathcal{G}$ has bounded expansion if and only if there exists a function f such that every r -shallow minor has chromatic number at most $f(r)$ [67, Proposition 5.5]. Moreover, graph classes with polynomial expansion have been characterized as those that admit strongly sublinear separators [36]. To prove this, Dvořák and Norin [36] showed that if a graph G admits strongly sublinear separators, then every r -shallow minor (for fixed $r \geq 1$) of G also admits strongly sublinear separators.

Before proceeding, note that the operation of taking a shallow minor of the product of a graph with a small complete graph has previously been studied within graph sparsity theory. In particular, Har-Peled and Quanrud [47] showed that $\nabla_r(G \circ K_\ell) \leq 5\ell^2(r+1)^2 \nabla_r(G)$ for every graph G (improving on an earlier result by Nešetřil and Ossona de Mendez [68]).

3. Shallow minors and graph products. This section explores various properties of shallow minors and graph products. For further structural results concerning graph products, see [34, 53] as well as the handbook [46].

3.1. Shortcut systems. We begin by examining the relationship between shallow minors and shortcut systems. Dujmović, Morin, and Wood [31] introduced shortcut systems as a way to prove product structure theorems for various non-minor-closed graph classes. A set $\mathcal{P}$ of paths in a graph G is a (k, d) -shortcut system if every path $P \in \mathcal{P}$ has length at most k and every vertex $v \in V(G)$ is an internal vertex for at most d paths in $\mathcal{P}$. Let $G^{\mathcal{P}}$ denote the supergraph of G obtained by adding the edge uv if $\mathcal{P}$ contains a (u, v) -path. Dujmović, Morin, and Wood [31] observed that k -planar graphs, d -map graphs, and several other classes can be described by applying shortcut systems to planar graphs. Using the following theorem, they deduced that these classes have a product structure.

THEOREM 4 (see [31]). *If $G \lesssim H \boxtimes P \boxtimes K_\ell$, for some graph H of treewidth at most t and $\mathcal{P}$ is a (k, d) -shortcut system for G , then $G^{\mathcal{P}} \lesssim J \boxtimes P \boxtimes K_{d\ell(k^3+3k)}$ for some graph J of treewidth at most $\binom{k+t}{t} - 1$ and some path P .*

In subsection 3.2, we adapt the proof of Theorem 4 to establish an analogous result in the more general setting of shallow minors.

Huynh and Wood [56] introduced the following variant of shortcut systems. A set $\mathcal{P}$ of paths in a graph G is a $(k, d)^*$ -shortcut system if every path in $\mathcal{P}$ has length at most k and for every $v \in V(G)$, if M_v is the set of vertices $u \in V(G)$ such that there exists a (u, w) -path in $\mathcal{P}$ in which v is an internal vertex, then $|M_v| \leq d$. Note that (k, d) -shortcut systems and $(k, d)^*$ -shortcut systems differ in how they count the number of paths that a vertex contributes to. Every $(k, d)^*$ -shortcut system is a $(k, \binom{d}{2})$ -shortcut system, and every (k, d) -shortcut system is a $(k, 2d)^*$ -shortcut system. Thus, $(k, d)^*$ -shortcut systems can give better bounds compared to (k, d) -shortcut systems. Shallow minors inherit this strength of $(k, d)^*$ -shortcut systems.

We show that graphs obtained by applying a shortcut system to a planar graph are k -gap planar and thus have linear crossing number. Later, we show that the class of fan-planar graphs, which has a shallow-minor structure, has superlinear crossing number and thus cannot be described by shortcut systems (see subsection 4.4).

LEMMA 5. *If $\mathcal{P}$ is a (k, d) -shortcut system of a planar graph G , then $G^{\mathcal{P}}$ is $((d-1)(k-1) + 2d)$ -gap planar.*

Proof. Assume that G is a plane graph. For some $\epsilon > \delta > 0$, for every vertex $x \in V(G)$ and edge $uv \in E(G)$, let

$$B_x := \{p \in \mathbb{R}^2 : \text{dist}_{\mathbb{R}^2}(p, x) \leq \epsilon\} \quad \text{and} \quad C_{uv} := \{p \in \mathbb{R}^2 : \text{dist}_{\mathbb{R}^2}(p, uv) \leq \delta\} \setminus (B_u \cup B_v).$$

Choosing ϵ and δ to be sufficiently small, we may assume that B_x, B_y, C_{ab} , and C_{uv} are nonempty and pairwise-disjoint for all $x, y \in V(G)$ and $ab, uv \in E(G)$. For each (x, y) -path $P = (x = w_0, w_1, \dots, w_{\ell-1}, w_{\ell} = y) \in \mathcal{P}$, embed the edge xy in the region

$$B_{w_0} \cup C_{w_0 w_1} \cup B_{w_1} \cup \dots \cup B_{w_{\ell-1}} \cup C_{w_{\ell-1} w_{\ell}} \cup B_{w_{\ell}}$$

so that for all $uv, xy \in E(G^{\mathcal{P}})$ with corresponding shortcuts $P_1, P_2 \in \mathcal{P}$, each crossing between uv and xy occurs in some B_w where w is an internal vertex of P_1 or P_2 , and uv and xy cross at most once in B_w (see Figure 1).

Let E_1 be the edges of G and $E_2 := E(G^{\mathcal{P}}) \setminus E_1$ be the new edges in $G^{\mathcal{P}}$. Since E_1 is a set of noncrossing edges, every crossing involves an edge from E_2 . If $uv \in E_1$ and $xy \in E_2$ cross, then charge the crossing to uv . Now suppose $x_1 y_1, x_2 y_2 \in E_2$ cross in B_w for some $w \in V(G)$. Let $P_1, P_2 \in \mathcal{P}$ be the shortcuts that respectively correspond to $x_1 y_1$ and $x_2 y_2$. If w is an internal vertex of P_1 , then charge the crossing to $x_2 y_2$. Otherwise w is an internal vertex of P_2 , in which case, charge the crossing to $x_1 y_1$.

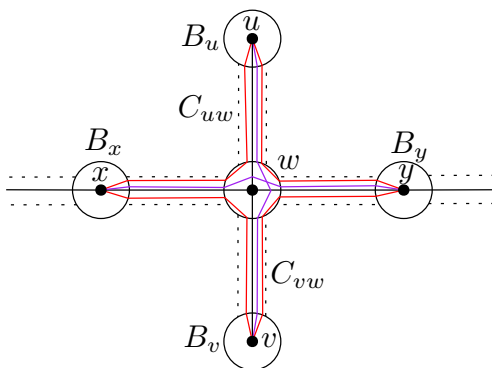


FIG. 1. Embedding $G^{\mathcal{P}}$ into the plane.

We now upper bound the number of crossings charged to an edge. For an edge $uv \in E_1$, if an edge $xy \in E_2$ crosses uv , then the (x, y) -path in $\mathcal{P}$ contains u or v as an internal vertex. As such, at most $2d$ crossings are charged to uv . Now consider an edge $x_1y_1 \in E_2$ with corresponding path $P_1 \in \mathcal{P}$. If an edge $x_2y_2 \in E_2$ with corresponding path $P_2 \in \mathcal{P}$ crosses x_1y_1 in B_w and the crossing is charged to x_1y_1 for some $w \in V(G)$, then w is an internal vertex of P_2 that is contained in $V(P_1)$. $\mathcal{P}$ has at most $2d$ paths that contain x_1 or y_1 as internal vertices. Furthermore, for each internal vertex $z \in V(P_1)$, at most $d - 1$ paths in $\mathcal{P} \setminus \{P_1\}$ contain z as an internal vertex. Since P_1 has at most $(k - 1)$ internal vertices, at most $(d - 1)(k - 1) + 2d$ crossings are charged to x_1y_1 . Therefore, $G^{\mathcal{P}}$ is $((d - 1)(k - 1) + 2d)$ -gap planar. $\square$

We now show that shallow minor structure subsumes shortcut systems.

LEMMA 6. *For every (k, d) -shortcut system $\mathcal{P}$ of a graph G , $G^{\mathcal{P}}$ is a $(\frac{k-1}{2})$ -shallow topological minor of $G \circ \overline{K}_{d+1}$.*

Proof. Assume that $V(K_{d+1}) = [d + 1]$ and that for each $uv \in E(G)$ there is a corresponding (u, v) -path in $\mathcal{P}$ with length 1. We will construct a topological minor of $G^{\mathcal{P}}$ in $G \circ \overline{K}_{d+1}$ where each vertex $v \in V(G^{\mathcal{P}})$ is mapped to $(v, 1) \in V(G \circ \overline{K}_{d+1})$. For each $w \in V(G)$, let M_w be the set of paths in $\mathcal{P}$ that contains w as an internal vertex. Let $\phi_w : M_w \rightarrow [2, d + 1]$ be an injective function. For each $v \in V(G)$, extend the domain and range of ϕ_v by letting $\phi_v(P) := 1$ for all paths $P \in \mathcal{P}$ with end-vertex v . For each path $P_{uv} = (u = w_0, w_1, \dots, w_{\ell-1}, w_{\ell} = v) \in \mathcal{P}$ where $\ell \leq k$, let $\tilde{P}_{uv}$ be the path in $G \circ \overline{K}_{d+1}$ defined by $V(\tilde{P}_{uv}) := \{(w_i, \phi_{w_i}(P_{uv})) : i \in [0, \ell]\}$ and $E(\tilde{P}_{uv}) := \{(w_i, \phi_{w_i}(P_{uv}))(w_{i+1}, \phi_{w_{i+1}}(P_{uv})) : i \in [0, \ell - 1]\}$. Let $\tilde{\mathcal{P}}$ be the set of such $\tilde{P}_{uv}$ paths.

We claim that $\tilde{\mathcal{P}}$ defines a $(\frac{k-1}{2})$ -shallow topological minor of $G^{\mathcal{P}}$ in $G \circ \overline{K}_{d+1}$ where each vertex $v \in V(G^{\mathcal{P}})$ is mapped to $(v, 1) \in V(G \circ \overline{K}_{d+1})$ and each edge $uv \in E(G^{\mathcal{P}})$ is mapped to $\tilde{P}_{uv} \in \tilde{\mathcal{P}}$. Let $uv \in E(G^{\mathcal{P}})$. Then there exists a path $P_{uv} \in \mathcal{P}$ with length at most k and end-vertices u and v . By construction, $\tilde{P}_{uv}$ is a path in $G \circ \overline{K}_{d+1}$ from $(u, 1)$ to $(v, 1)$ with length at most k . Thus, it suffices to show that the paths in $\tilde{\mathcal{P}}$ are internally disjoint. The internal vertices of each path $\tilde{P}_{uv} \in \tilde{\mathcal{P}}$ are of the form $(w, \phi_w(P_{uv}))$ where $\phi_w(P_{uv}) \in [2, d + 1]$. Suppose there is another path $\tilde{Q}_{xy} \in \tilde{\mathcal{P}}$ for which $(w, \phi_w(P_{uv}))$ is an internal vertex of $\tilde{Q}_{xy}$. Then $\tilde{P}_{uv} = \tilde{Q}_{xy}$ since ϕ_w is injective. As such, the paths in $\tilde{\mathcal{P}}$ are internally disjoint, as required. $\square$

3.2. Key tool. Having shown that shallow minors subsume shortcut systems, we now show that shallow minors inherit product structure.

THEOREM 7. *For all graphs H and L , if a graph G is an r -shallow minor of $H \boxtimes L \boxtimes K_{\ell}$ where H has treewidth at most t and $\Delta(L^r) \leq k$, then $G \lesssim J \boxtimes L^{2r+1} \boxtimes K_{\ell(k+1)}$ for some graph J with treewidth at most $\binom{2r+1+t}{t} - 1$.*

To prove Theorem 7, we use the language of H -partitions from Dujmović et al. [28]. Let G be a graph. A *partition* of G is a collection $\mathcal{Y}$ of sets of vertices in G such that each vertex of G is in exactly one part of $\mathcal{Y}$. The *quotient* of $\mathcal{Y}$ (with respect to G) is the graph, denoted by $G/\mathcal{Y}$, with vertex set $\mathcal{Y}$ where distinct parts $A, B \in \mathcal{Y}$ are adjacent in $G/\mathcal{Y}$ if and only if some vertex in A is adjacent in G to some vertex in B . An H -*partition* of G is a partition $\mathcal{Y} = (Y_x : x \in V(H))$ indexed by the nodes of H such that $G/\mathcal{Y} \lesssim H$. Note that $\mathcal{Y}$ is allowed to contain some empty parts. For our purposes, we require the following extension of H -partitions. An (H, L) -*partition* $(\mathcal{Y}, \mathcal{Z})$ consists of an H -partition $\mathcal{Y}$ and an L -partition $\mathcal{Z}$ of G for some graphs H and L . The *width* of $(\mathcal{Y}, \mathcal{Z})$ is $\max\{|Y_y \cap Z_z| : y \in V(H), z \in V(L)\}$.

The following generalizes an observation by Dujmović et al. [28].

Observation 1. For all graphs H and L , a graph G is contained in $H \boxtimes L \boxtimes K_\ell$ if and only if G has an (H, L) -partition with width at most ℓ .

For a rooted tree (T, x_0) , we say that a node $a \in V(T)$ is a T -ancestor of $x \in V(T)$ (and x is a T -descendent of a) if a is contained in the path in T from x_0 to x . If in addition $a \neq x$, then a is a *strict* T -ancestor of x . Note that x is a T -ancestor and a T -descendent of itself.

The proof of Theorem 7 is an adaptation of the proof of Theorem 4 [31, Theorem 9]. We make use of the following well-known normalization lemma (see [31, Lemma 2] for a proof).

LEMMA 8. For every graph H of treewidth t , there is a rooted tree (T, x_0) with $V(T) = V(H)$ and a width- t tree-decomposition $(T, \{W_x : x \in V(T)\})$ of H that has the following additional properties:

- (T1) for each node $x \in V(H)$, the subtree $T[x] := T[\{y \in V(T) : x \in W_y\}]$ is rooted at x ; and consequently
- (T2) for each edge $xy \in E(H)$, one of x or y is a T -ancestor of the other.

We now prove our main technical lemma which, together with Observation 1, implies Theorem 7.

LEMMA 9. Let G be a graph having an (H, L) -partition with width ℓ in which H has treewidth at most t and $\Delta(L^r) \leq k$. Then every r -shallow minor G' of G has a (J, L^{2r+1}) -partition with width at most $\ell(k+1)$ where the graph J has treewidth at most $\binom{2r+1+t}{t} - 1$.

Proof. Let μ be an r -shallow model of G' in G . Assume that $V(G') \subseteq V(G)$ and that u is a center of $\mu(u)$ for each $u \in V(G')$. Let $\mathcal{Y} := (Y_x : x \in V(H))$ and $\mathcal{Z} := (Z_z : z \in V(L))$ respectively be an H -partition and an L -partition of G , where $(\mathcal{Y}, \mathcal{Z})$ has width at most ℓ . Let $\mathcal{B} = (B_x : x \in V(T))$ be a tree-decomposition of H that satisfies the conditions of Lemma 8. Note that $V(T) = V(H)$. Let x_0 denote the root of T . For a vertex $u \in V(G')$, let $X_u := \{x \in V(H) : V(\mu(u)) \cap Y_x \neq \emptyset\}$; that is, X_u is the set of nodes in H that indexes a part in a $\mathcal{Y}$ which contains a vertex from $\mu(u)$.

Claim 1. For every $u \in V(G')$, there exists a node $a(u) \in X_u$ such that $a(u)$ is a T -ancestor of every node in X_u .

Proof. Since $\mathcal{Y}$ is a partition of G and $\mu(u)$ is connected, $H[X_u]$ is connected. By the transitivity of the T -ancestor relationship, (T2), there exists a node $a(u) \in X_u$ such that $a(u)$ is a T -ancestor of every node in X_u . $\square$

Note that $a(u)$ is the vertex in X_u that is closest (in T) to x_0 . We now use $a(u)$ to define a partition of G' . For each $x \in V(T)$, define $S_x := \{u \in V(G') : a(u) = x\}$. Observe that $\mathcal{S} := (S_x : x \in V(T))$ is a partition of $V(G')$. Let $J := G'/\mathcal{S}$ denote the resulting quotient graph, and let $V(J) \subseteq V(T)$ where each $x \in V(J)$ is obtained by identifying S_x in G' . For each $z \in V(L)$, let $Z'_z := Z_z \cap V(G')$ and $\mathcal{Z}' := (Z'_z : z \in V(L))$.

From here on in, we need to show (i) that $\mathcal{S}$ has small width with respect to $\mathcal{Z}'$, (ii) that J has small treewidth, and (iii) that $G'/\mathcal{Z}'$ is contained in L^{2r+1} . The next claim demonstrates (i).

Claim 2. $|S_x \cap Z'_z| \leq \ell(k+1)$ for all $x \in V(J)$ and $z \in V(L)$.

Proof. Let $u \in S_x \cap Z'_z$ and $W := \{j \in V(L) : \text{dist}_L(z, j) \leq r\}$. By definition, $|W| \leq k + 1$. Since $a(u) = x$, there is a vertex $w \in V(\mu(u))$ that is contained in the part Y_x (recall that Y_x is the part in the H -partition of G that is indexed by x). Since $\text{dist}_G(u, w) \leq r$ and L is a partition of G , $w \in \bigcup_{j \in W} Z_j$. As such, $w \in Y_x \cap (\bigcup_{j \in W} Z_j)$. Therefore $|S_x \cap Z'_z| \leq \ell(k + 1)$ since each vertex in $Y_x \cap (\bigcup_{j \in W} Z_j)$ can contribute at most one vertex to $S_x \cap Z'_z$ and $|Y_x \cap (\bigcup_{j \in W} Z'_j)| \leq \ell(k + 1)$. $\square$

The following claim will be useful in bounding the treewidth of J .

Claim 3. For each edge $xy \in E(J)$, one of x or y is a T -ancestor of the other.

Proof. Our goal is to show that there exists a node $w \in V(T)$ such that x and y are both T -ancestors of w . This implies that one of x or y is a T -ancestor of the other.

Since $xy \in E(J)$, G' contains an edge uv with $u \in S_x$ and $v \in S_y$. By the definition of S , it follows that $a(u) = x$ and $a(v) = y$. By Claim 1, x is a T -ancestor of every node in X_u and y is a T -ancestor of every node in X_v . Since $uv \in E(G')$, there is an edge $\tilde{u}\tilde{v} \in E(G)$ where $\tilde{u} \in V(\mu(u))$ and $\tilde{v} \in V(\mu(v))$. Let $\tilde{x} \in V(T)$ and $\tilde{y} \in V(T)$ respectively be the nodes in T such that $\tilde{u} \in Y_{\tilde{x}}$ and $\tilde{v} \in Y_{\tilde{y}}$. Since $\tilde{u} \in V(\mu(u))$ and $\tilde{v} \in V(\mu(v))$, it follows that $\tilde{x} \in X_u$ and $\tilde{y} \in X_v$. So x is a T -ancestor of $\tilde{x}$, and y is a T -ancestor of $\tilde{y}$. As such, if $\tilde{x} = \tilde{y}$, then we are done by setting $w := \tilde{x}$. Otherwise, $\tilde{x}\tilde{y} \in E(H)$ since $\tilde{u}\tilde{v} \in E(G)$. By (T2), either $\tilde{x}$ or $\tilde{y}$ is a T -ancestor of the other. Without loss of generality, assume that $\tilde{x}$ is a T -ancestor of $\tilde{y}$. Then by setting $w := \tilde{y}$, it follows that x and y are both T -ancestors of w , as required. $\square$

We now show that J has small treewidth.

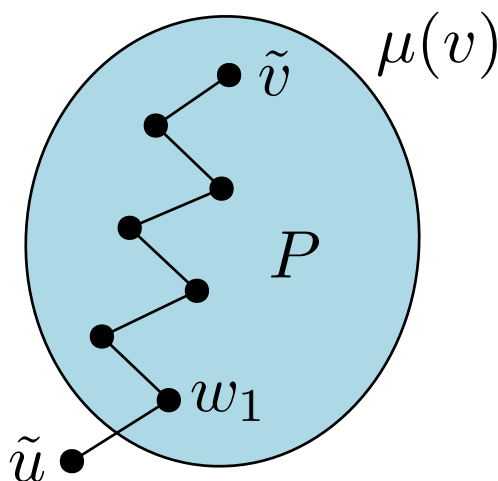
Claim 4. J has a tree-decomposition in which every bag has size at most $\binom{2r+1+t}{t}$.

Proof. We will define a tree-decomposition of J with small width that is indexed by the tree T . For each node x of T , let C_x be a bag that contains x as well as every T -ancestor a of x such that J contains an edge ax' where x' is a T -descendant of x . Clearly, every vertex is in a bag. Claim 3 ensures that every edge is in a bag. The connectedness of $T[a] := T[\{x \in V(T) : a \in C_x\}]$ follows from the fact that for every edge $x'a \in E(J)$ where x' is a T -descendant of a , every node x on the (x', a) -path in T is a node in $T[a]$. As such, $(C_x : x \in V(T))$ defines a tree-decomposition of J . It remains to bound the size of each bag C_x .

Let H^+ denote the supergraph of H with vertex set $V(H)$ in which $xy \in E(H^+)$ if and only if $x, y \in B$ for some bag $B \in \mathcal{B}$. Let $\vec{H}^+$ be the directed graph obtained by directing each edge $xy \in E(H^+)$ from its T -descendant x toward its T -ancestor y . Our goal is to show that for every $x \in V(T)$, if $a \in C_x$, then there is a directed path of length at most $2r + 1$ in $\vec{H}^+$ from x to a . Once we have shown that, we may then appeal to [73, Lemma 13], which states that the number of nodes in $\vec{H}^+$ that can be reached from any node x by a directed path of length at most $2r + 1$ is at most $\binom{2r+1+t}{t}$, thus implying our claim.

Consider an arbitrary node $x \in V(T)$ where $x_0, \dots, x_h$ is the path from the root x_0 of T to $x_h := x$. To simplify our notation, let Y_i and S_i be shorthand for Y_{x_i} and S_{x_i} , respectively. For each $i \in [0, h]$, let $V_i := \bigcup (Y_y : y \text{ is a } T\text{-descendant of } x_i)$; that is, V_i is the set of vertices in G that are contained in a part that is indexed by a T -descendant of x_i . Observe that $V_i \subseteq V_j$ whenever $j \geq i$.

Let $x_\delta \in C_x$. We now work toward showing that there is a directed path of length at most $2r + 1$ in $\vec{H}^+$ from x to x_δ . Since $x_\delta \in C_x$, it is because x_δ is adjacent in J to some T -descendant of x or $x_\delta = x$. In the case when $x_\delta = x$, there is trivially a directed path in $\vec{H}^+$ from x to x_δ of length at most $2r + 1$. So we may assume there is a T -descendant x' of x such that $x_\delta x' \in E(J)$. So G' contains an edge uv with

FIG. 2. Path P from $\tilde{u}$ to $\tilde{v}$.

$u \in S_{x'}$ and $v \in S_\delta$. Since $a(v) = x_\delta$, there exists $\tilde{v} \in V(\mu(v)) \cap Y_\delta$. Let $\tilde{u} \in V(\mu(u))$ be such that there exists a path $P = (\tilde{u} = w_0, w_1, \dots, w_p = \tilde{v})$ in $G[V(\mu(v)) \cup \{\tilde{u}\}]$ where $p \leq 2r + 1$ (see Figure 2).

For each $i \in [0, p]$, let $s_i := \max \{\ell \in [0, h] : w_0, \dots, w_i \subseteq V_\ell\}$, and let $a_i := x_{s_i}$. That is, a_i is the furthest vertex from the root on the (x_0, x_h) -path such that $w_0, \dots, w_i \subseteq V_{a_i}$. Then $s_0, \dots, s_p$ is a nonincreasing sequence and $a_0, \dots, a_p$ is a sequence of nodes of T whose distance from the root x_0 is nonincreasing.

We claim that $a_0 = x_h$. Since $a(u) = x'$ which is a T -descendant of x_h , it follows by Claim 1 that $\tilde{u} \in V_h$. Since h is trivially the maximum of $\{0, 1, \dots, h\}$, $a_0 = x_h$ as required.

We claim that $a_p = x_\delta$. Since $a(u)$ and $a(v)$ are T -descendant of x_δ , by Claim 1, $w_0, w_1, \dots, w_p \subseteq V_\delta$ and so a_p is a T -descendant of x_δ . Since $\tilde{v} \in Y_\delta$, it follows that a_p is a T -ancestor of x_δ . Hence $a_p = x_\delta$ as required.

We claim that $a_0, \dots, a_p$ is a lazy walk¹ in H^+ . Suppose that $a_i \neq a_{i+1}$ for some $i \in [0, p - 1]$. Then $w_i \in V_{a_i}$ and $w_{i+1} \notin V_{a_i}$. Let b_i and c_i be the unique nodes where $w_i \in Y_{b_i}$ and $w_{i+1} \in Y_{c_i}$. By definition of V_{a_i} , b_i is a T -descendant of a_i . Now $b_i c_i \in E(H)$ since $w_i w_{i+1} \in E(G)$, and thus by (T2), c_i is a strict T -ancestor of a_i . Since $w_1, \dots, w_i, w_{i+1} \in V_{c_i}$ and $w_{i+1} \in Y_{c_i}$, it follows that $a_{i+1} = c_i$. By (T1), $a_{i+1} \in B_{a_{i+1}}$ and $a_{i+1} \in B_{b_i}$. Since a_i is on the path from b_i to a_{i+1} in T , this implies $a_{i+1} \in B_{a_i}$. Therefore, $a_i a_{i+1} \in E(H^+)$ as required. So by removing repeated vertices from this lazy walk, we obtain a path of length at most $2r + 1$ in the directed graph $\vec{H}^+$ from x to x_δ , as required. $\square$

To finish the proof, note that $\mathcal{Z}'$ may not be an L -partition of G' . In particular, for every edge $vw \in E(G')$ with $v \in Z'_i$ and $w \in Z'_j$, we have $\text{dist}_L(i, j) \leq 2r + 1$. Thus, by indexing $\mathcal{Z}'$ by L^{2r+1} instead of L , we obtain a valid partition of G' . Therefore, G' has a (J, L^{2r+1}) -partition with width at most $\ell(k + 1)$ where $\text{tw}(J) \leq \binom{2r+t}{t} - 1$. $\square$

Recall that the row treewidth $\text{rtw}(G)$ of a graph G is the minimum treewidth of a graph H such that $G \subseteq H \boxtimes P$ for some path P . Since $\Delta(P^r) \leq 2r$ and $P^{2r+1} \subseteq P \boxtimes K_{2r+1}$, we have the following consequence of Theorem 7.

¹A *lazy walk* in a graph G is a sequence $(v_1, \dots, v_n)$ of vertices such that v_i and v_{i+1} are adjacent or equal for each $i \in [n - 1]$.

THEOREM 10. *If G is an r -shallow minor of $H \boxtimes P \boxtimes K_\ell$ where H has treewidth at most t and P is a path, then $G \preceq J \boxtimes P \boxtimes K_{\ell(2r+1)^2}$, where J has treewidth at most $\binom{2r+1+t}{t} - 1$, and thus $\text{rtw}(G) \leq \binom{2r+1+t}{t} \ell(2r+1)^2 - 1$.*

Theorem 10 is used in section 4 to establish product structure theorems for beyond-planar graphs.

3.3. Stack and queue layouts. Heath, Leighton, and Rosenberg [49, 50] introduced stack and queue layouts as a way to measure the power of stacks and queues to represent graphs. Let G be a graph and $\preceq$ be a total order on $V(G)$. Two disjoint edges $vw, xy \in E(G)$ with $v \prec w$ and $x \prec y$:

- *cross* with respect to $\preceq$ if $v \prec x \prec w \prec y$ or $x \prec v \prec y \prec w$;
- *nest* with respect to $\preceq$ if $v \prec x \prec y \prec w$ or $x \prec v \prec w \prec y$; and
- *overlap* with respect to $\preceq$ if $v = x$ or $w = y$.

Consider a function $\varphi : E(G) \rightarrow [k]$ for some integer $k \geq 1$. Then $(\preceq, \varphi)$ is a k -stack layout of G if vw and xy do not cross for all edges $vw, xy \in E(G)$ with $\varphi(vw) = \varphi(xy)$. Similarly, $(\preceq, \varphi)$ is a k -queue layout of G if vw and xy do not nest for all edges $vw, xy \in E(G)$ with $\varphi(vw) = \varphi(xy)$. If vw and xy neither nest nor overlap for all edges $vw, xy \in E(G)$ with $\varphi(vw) = \varphi(xy)$, then $(\preceq, \varphi)$ is a *strict k -queue layout* of G . The minimum integer $s \geq 0$ for which G has an s -stack layout is the *stack-number*, $\text{sn}(G)$, of G . The minimum integer $q \geq 0$ for which G has a q -queue layout is the *queue-number*, $\text{qn}(G)$, of G . The minimum integer $q \geq 0$ for which G has a strict q -queue layout is the *strict-queue-number*, $\text{sqn}(G)$, of G . Stack layouts are equivalent to book embeddings (first defined by Ollmann [70]) and stack-number is also known as *page-number*, *book-thickness*, and *fixed outer-thickness*.

Wood [84] showed that $\text{qn}(G \boxtimes H) \leq 2\text{sqn}(H) \cdot \text{qn}(G) + \text{sqn}(H) + \text{qn}(G)$ for all graphs G and H . Observe that $1 \preceq 2 \preceq \dots \preceq \ell$ together with $\phi(ij) = |i - j|$ defines a strict $(\ell - 1)$ -queue layout of K_ℓ ; thus $\text{sqn}(K_\ell) \leq \ell - 1$. Hence, for every graph G ,

$$(\star) \quad \text{qn}(G \boxtimes K_\ell) \leq (2\ell - 1) \text{qn}(G) + \ell - 1.$$

Dujmović, Morin, and Wood [29] proved that graphs of bounded treewidth have bounded queue-number. The best known bound is $\text{qn}(G) \leq 2^{\text{tw}(G)} - 1$, due to Wiechert [83]. Using techniques similar to the result of Wood [84], Dujmović et al. [28] proved the following.

LEMMA 11 (see [28]). *For every graph H and every path P , if $G \preceq H \boxtimes P \boxtimes K_\ell$ then*

$$\text{qn}(G) \leq 3\ell 2^{\text{tw}(H)} + \lfloor \frac{3}{2}\ell \rfloor.$$

Using Theorem 2 and Lemma 11, Dujmović et al. [28] showed that planar graphs have queue-number at most 49. This result answered an important open problem of Heath, Leighton, and Rosenberg [49] asking whether planar graphs have bounded queue-number. Refining the proof in [28], Bekos, Gronemann, and Raftopoulou [6] strengthened this upper bound to 42.

In contrast to queue layouts, stack layouts are often not well-behaved under products. Dujmović et al. [25] provided the first such example by showing that $\{P \boxtimes P \boxtimes S : P \text{ is a path, } S \text{ is a star}\}$ has unbounded stack-number (despite $P \boxtimes P$ and $P \boxtimes S$ having bounded stack-number). Similarly, Eppstein et al. [40] showed that $\{P \boxtimes P \boxtimes P : P \text{ is a path}\}$ has unbounded stack-number. Most recently, Leung [65] showed that $\{T \boxtimes P : T \text{ is a tree, } P \text{ is a path}\}$ has unbounded stack-number. (In each of the above results, the authors actually proved slightly stronger statements.) In the

positive direction, Pupyrev [74] showed that for every path P , $H \boxtimes P$ has bounded stack-number whenever the graph H has bounded pathwidth or is bipartite with bounded maximum degree and bounded treewidth.

Now consider the relationship between shallow minors and stack and queue layouts. For stack layouts, Dujmović et al. [25] described a family of graphs $(G_n)_{n \in \mathbb{N}}$ with unbounded stack-number where for every $n \in \mathbb{N}$, the 5-subdivision of G_n has stack-number at most 3. As such, stack-number is not “well-behaved” under shallow topological minors. This result has been recently amplified to show that stack-number is not even “well-behaved” under small minors; that is, there exists a graph class $\mathcal{G}$ with bounded stack-number such that the graph class that consists of 2-small minors of graphs in $\mathcal{G}$ has unbounded stack-number [40]. On the other hand, queue-number is well-behaved: Dujmović and Wood [32] showed that if H is an r -shallow topological minor of a graph G , then $\text{qn}(H) \leq \frac{1}{2}(2\text{qn}(G) + 1)^{4r} - 1$. The next lemma generalizes this result to the setting of shallow minors.

LEMMA 12. *For every graph G , and for every r -shallow minor H of G ,*

$$\text{qn}(H) \leq (2r + 1)(2\text{qn}(G))^{2r+1}.$$

Proof. Let H be an r -shallow minor of G where $V(H) \subseteq V(G)$ and let μ be an r -shallow model of H in G where v is a center of $\mu(v)$ for each $v \in V(H)$.

Let $(\preceq_G, \psi_G)$ be a $\text{qn}(G)$ -queue layout of G . For a directed edge $\overrightarrow{xy} \in E(G)$, we say that it is *forward* if $x \prec_G y$. Let $\preceq_H$ be the induced order of $V(H)$; that is, if $u \prec_G v$ for some $u, v \in V(H)$, then $u \prec_H v$. Let $uv \in E(H)$ where $u \prec_H v$ and $P_{uv} = (u = w_0, \dots, w_j = v)$ be a (u, v) -path in G where $j \leq 2r + 1$ and $P_{uv} \subseteq \mu(u) \cup \mu(v)$. For each $i \in [j]$, direct the edge $\overrightarrow{w_{i-1}w_i}$. Record the following for the queue assignment $\psi_H(uv)$: the length of P_{uv} , $\psi_G(w_{i-1}w_i)$ for each $i \in [j]$, and $\{i \in [j] : \overrightarrow{w_{i-1}w_i} \text{ is forward}\}$. Thus, there are at most $(2r + 1) \times \text{qn}(G)^{2r+1} \times 2^{2r+1}$ queue assignments of the edges of H .

We claim that $(\preceq_H, \psi_H)$ is a $(2r(2\text{qn}(G))^{2r})$ -queue layout of G . Suppose there exists $uv, xy \in E(H)$ such that $\psi_H(uv) = \psi_H(xy)$ with corresponding paths $P_{uv} = (u = w_0, \dots, w_j = v)$ and $P_{xy} = (x = z_0, \dots, z_l = y)$ in G . Note that P_{uv} and P_{xy} have the same length because $\psi_H(uv) = \psi_H(xy)$. Now if uv and xy share a common end-vertex, then they do not nest. So assume that $\{u, v\} \cap \{x, y\} = \emptyset$ and thus $V(P_{uv}) \cap V(P_{xy}) = \emptyset$. Moreover, assume that $u \prec_H v$ and $x \prec_H y$ and $u \prec_H x$. We claim that $w_i \prec_G z_i$ for all $i \in [0, j]$. For $i = 0$, the claim holds by assumption. By induction, we may assume that $w_{i-1} \prec_G z_{i-1}$. Now since $\psi_H(uv) = \psi_H(xy)$, we have $\psi_G(w_{i-1}w_i) = \psi_G(z_{i-1}z_i)$ and without loss of generality, assume that $\overrightarrow{w_{i-1}w_i}$ and $\overrightarrow{z_{i-1}z_i}$ are both forward. Now if $z_i \prec_G w_i$, then $w_{i-1} \prec_G z_{i-1} \prec_G z_i \prec_G w_i$. Since $\{w_{i-1}, w_i\} \cap \{z_{i-1}, z_i\} = \emptyset$, it follows that $w_{i-1}w_i$ and $z_{i-1}z_i$ nest, a contradiction. Therefore, $v \prec_H y$ and hence xy and uv do not nest, as required. $\square$

For the beyond-planar graphs defined in section 4, we can use their product structure (with Lemma 11) or their shallow-minor structure (with Lemma 12) to prove that they have bounded queue-number. In some cases, the product structure gives better bounds, whereas in other cases, the shallow-minor structure gives better bounds.

3.4. Nonrepetitive colorings. Nonrepetitive colorings were introduced by Alon et al. [2] and have since been widely studied (see the survey [85]). A vertex c -coloring ϕ of a graph G is *nonrepetitive* if for every path $v_1, \dots, v_{2h}$ in G , there exists $i \in [h]$ such that $\phi(v_i) \neq \phi(v_{i+h})$. The *nonrepetitive chromatic number*, $\pi(G)$, is the minimum integer $c \geq 0$ such that G has a nonrepetitive c -coloring.

Kündgen and Pelsmayer [63] showed that $\pi(G) \leq 4^{\text{tw}(G)}$ for every graph G . Building upon this result, Dujmović et al. [27] proved the following.

LEMMA 13 (see [27]). *If $G \subseteq H \boxtimes P \boxtimes K_\ell$, then $\pi(G) \leq \ell 4^{\text{tw}(H)+1}$.*

Dujmović et al. [27] used Theorem 2 and Lemma 13 to show that planar graphs have bounded nonrepetitive chromatic number, resolving a long-standing conjecture of Alon et al. [2].

We highlight the following open problem concerning nonrepetitive chromatic number. Nešetřil, Ossona de Mendez, and Wood [69] showed that nonrepetitive chromatic number is “well-behaved” under shallow topological minors. We ask whether this can be generalized to shallow minors. More formally, does there exist a function f such that $\pi(H) \leq f(r, \pi(G))$ whenever H is an r -shallow minor of G ?

3.5. Centered colorings. Nešetřil and Ossona de Mendez [68] introduced the following definition. A vertex c -coloring ϕ of a graph G is p -centered if, for every connected subgraph $X \subseteq G$, $|\{\phi(v) : v \in V(X)\}| > p$ or there exists some $v \in V(X)$ such that $\phi(v) \neq \phi(w)$ for every $w \in V(X) \setminus \{v\}$. That is, X receives more than p colors or some vertex in X receives a unique color. For an integer $p \geq 1$, the p -centered chromatic number, $\chi_p(G)$, is the minimum integer $c \geq 0$ such that G has a p -centered c -coloring. Centered colorings are important within graph sparsity theory as they characterize graph classes with bounded expansion [68].

Debski et al. [20] established that $\chi_p(G \boxtimes H) \leq \chi_p(G)\chi(H^p)$ for all graphs G and H . Pilipczuk and Siebertz [73, Lemma 15] showed that $\chi_p(G) \leq \binom{p+t}{t}$ for every graph G with treewidth at most t . The next result follows.

LEMMA 14 (see [20, 31]). *If $G \subseteq H \boxtimes P \boxtimes K_\ell$ and $\text{tw}(H) \leq t$, then*

$$\chi_p(G) \leq \ell(p+1)\chi_p(H) \leq \ell(p+1)\binom{p+t}{t}.$$

3.6. Layered treewidth. Layered tree-decompositions were introduced by Dujmović, Morin, and Wood [30] as a precursor to graph product structure theory. A *layering* of a graph G is an ordered partition $\mathcal{L} := (L_0, L_1, \dots)$ of $V(G)$ such that for every edge $vw \in E(G)$, if $v \in L_i$ and $w \in L_j$, then $|i - j| \leq 1$. A *layered tree-decomposition* $(\mathcal{L}, \mathcal{T})$ consists of a layering $\mathcal{L}$ and a tree-decomposition $\mathcal{T} = (T, \mathcal{B})$ of G . The *layered width* of $(\mathcal{L}, \mathcal{T})$ is $\max\{|L \cap B| : L \in \mathcal{L}, B \in \mathcal{B}\}$. The *layered treewidth*, $\text{ltw}(G)$, of G is the minimum layered width of any layered tree-decomposition of G . It is known that $\text{ltw}(G) \leq \text{rtw}(G) + 1$ for every graph G [28] but for every integer $k \geq 1$, there exists a graph G with $\text{ltw}(G) = 1$ and $\text{rtw}(G) \geq k$ [14]. Dujmović, Morin, and Wood [30] showed that every graph G with Euler genus g has $\text{ltw}(G) \leq 2g + 3$. In addition, Dujmović, Morin, and Wood [30, Lemma 9] proved the following.

LEMMA 15 (see [30]). *For every graph G , and for every r -shallow minor H of $G \boxtimes K_\ell$,*

$$\text{ltw}(H) \leq \ell(4r + 1)\text{ltw}(G).$$

We mention several applications of layered treewidth. For strong coloring numbers (see subsection 3.7), van den Heuvel and Wood [82] proved that $\text{scol}_s(G) \leq \text{ltw}(G)(2s + 1)$ for every graph G . The *boxicity* of a graph G , $\text{box}(G)$, is the minimum integer $d \geq 1$, such that G is the intersection graph of axis-aligned boxes in $\mathbb{R}^d$. Thomassen [79] established that planar graphs have boxicity at most 3. Scott and Wood [78] showed that $\text{box}(G) \leq 6\text{ltw}(G) + 4$ for every graph G . A closer inspection of their proof gives the following bound.

LEMMA 16 (see [78]). $\text{box}(G) \leq 6\text{ltw}(G) + 3$ for every graph G .

At this stage, it is worth comparing layered treewidth to row treewidth. For many applications, row treewidth has superseded layered treewidth by giving qualitatively stronger bounds. For example, while layered treewidth implies that n -vertex planar graphs have queue-number $O(\log(n))$ [30], row treewidth implies that planar graphs have bounded queue-number [28].² Nevertheless, there are several applications (in particular, boxicity and strong coloring numbers) where row treewidth has not been shown to give stronger bounds than layered treewidth. Moreover, for many beyond-planar graphs, we obtain much better bounds for layered treewidth than we do for row treewidth (see section 4). For example, using Lemma 15 and Theorem 10, we show that fan-planar graphs have row treewidth at most 1619 and layered treewidth at most 45 (see subsection 4.4). As such, we obtain significantly stronger bounds for the boxicity and strong coloring numbers of fan-planar graphs using layered treewidth than via row treewidth. This highlights the value of layered treewidth, especially for beyond-planar graphs.

3.7. Generalized coloring numbers. Kierstead and Yang [60] introduced the following definitions. For a graph G , a total order $\preceq$ of $V(G)$, a vertex $v \in V(G)$, and an integer $s \geq 1$, let $R(G, \preceq, v, s)$ be the set of vertices $w \in V(G)$ for which there is a path $v = w_0, w_1, \dots, w_{s'} = w$ of length $s' \in [0, s]$ such that $w \preceq v$ and $v \prec w_i$ for all $i \in [s' - 1]$, and let $Q(G, \preceq, v, s)$ be the set of vertices $w \in V(G)$ for which there is a path $v = w_0, w_1, \dots, w_{s'} = w$ of length $s' \in [0, s]$ such that $w \preceq v$ and $w \prec w_i$ for all $i \in [s' - 1]$. For a graph G and integer $s \geq 1$, the s -strong coloring number, $\text{scol}_s(G)$, is the minimum integer such that there is a total order $\preceq$ of $V(G)$ with $|R(G, \preceq, v, s)| \leq \text{scol}_s(G)$ for every vertex v of G . Likewise, the s -weak coloring number, $\text{wcol}_s(G)$, is the minimum integer such that there is a total order $\preceq$ of $V(G)$ with $|Q(G, \preceq, v, s)| \leq \text{wcol}_s(G)$ for every vertex v of G .

Coloring numbers provide upper bounds on several graph parameters of interest. First note that $\text{scol}_1(G) = \text{wcol}_1(G)$ which equals the degeneracy of G plus 1, implying $\chi(G) \leq \text{scol}_1(G)$. A proper graph coloring is *acyclic* if the union of any two color classes induces a forest; that is, every cycle is assigned at least three colors. The *acyclic chromatic number* $\chi_a(G)$ of a graph G is the minimum integer k such that G has an acyclic k -coloring. Kierstead and Yang [60] proved that $\chi_a(G) \leq \text{scol}_2(G)$ for every graph G . Other parameters that can be bounded by strong and weak coloring numbers include game chromatic number [59, 60], Ramsey numbers [16], oriented chromatic number [62], arrangeability [16], boxicity [43], odd chromatic number [51], and conflict-free chromatic number [51].

Another attractive aspect of strong coloring numbers is that they interpolate between degeneracy and treewidth [44]. As previously noted, $\text{scol}_1(G)$ equals the degeneracy of G plus 1. At the other extreme, Grohe et al. [44] showed that $\text{scol}_s(G) \leq \text{tw}(G) + 1$ for every integer $s \geq 1$, and indeed $\text{scol}_s(G) \rightarrow \text{tw}(G) + 1$ as $s \rightarrow \infty$.

Coloring numbers are important because they characterize bounded expansion [86] and nowhere dense classes [44] and have several algorithmic applications [35, 45].

We prove the following.

THEOREM 17. *For every graph G , for every r -shallow minor H of $G \boxtimes K_\ell$, and for every integer $s \geq 1$,*

²It is open whether queue-number is bounded by layered treewidth. In fact, Bose et al. [14] showed that graphs with bounded layered treewidth have bounded queue-number if and only if graphs with layered treewidth 1 have bounded queue-number.

$$\text{scol}_s(H) \leq \ell \text{scol}_{2rs+2r+s}(G) \quad \text{and} \quad \text{wcol}_s(H) \leq \ell \text{wcol}_{2rs+2r+s}(G).$$

A graph class $\mathcal{G}$ has *linear strong coloring numbers* if there is a constant $c > 0$ such that $\text{scol}_s(G) \leq cs$ for every $G \in \mathcal{G}$ and for every integer $s \geq 1$. Say $\mathcal{G}$ has linear strong coloring numbers. Let $\mathcal{G}_r^\ell$ be the class of r -shallow minors of graphs in $\mathcal{G} \boxtimes K_\ell$. Then for all graphs $H \in \mathcal{G}_r^\ell$ and integer $s \geq 1$,

$$\text{scol}_s(H) \leq \ell \text{scol}_{2rs+2r+s}(G) \leq c\ell(2rs + 2r + s) \leq c\ell(4r + 1)s.$$

So $\mathcal{G}_r^\ell$ has linear strong coloring numbers (with corresponding constant $c\ell(4r + 1)$). In particular, for graphs with bounded Euler genus, van den Heuvel et al. [81] proved that $\text{scol}_s(G) \leq (4g + 5)s + 2g + 1$ and $\text{wcol}_s(G) \leq (2g + \binom{s+2}{2})(2s + 1)$ for every integer $s \geq 1$ and graph G with Euler genus g . Thus Theorem 17 implies the following.

THEOREM 18. *For every graph G with Euler genus g , if H is an r -shallow minor of $G \boxtimes K_\ell$ then*

$$\begin{aligned} \text{scol}_s(H) &\leq \ell((4g + 5)(2rs + 2r + s) + 2g + 1) \quad \text{and} \\ \text{wcol}_s(H) &\leq \ell(2g + \binom{(2r+1)s+2r+2}{2})((4r + 2)s + 4r + 1). \end{aligned}$$

We use Theorem 18 in section 4 to demonstrate that several beyond-planar graph classes have linear strong coloring numbers and cubic weak coloring numbers. Layered treewidth also implies linear strong coloring numbers for these graph classes [82]. A strength of our approach is that if the upper bound for the strong or weak coloring numbers for graphs with bounded Euler genus is improved, then this would improve the bounds for the beyond-planar graphs in section 4. For example, Joret and Micek [57] conjectured that planar graphs have s -weak coloring numbers $\Theta(s^2 \log(s))$. If true, this would imply that the beyond-planar graphs in section 4 have s -weak coloring numbers $\Theta(s^2 \log(s))$.

We now work toward proving Theorem 17. Sergey Norin (see [42]) observed that every r -shallow minor of a graph G has average degree at most $2\text{scol}_{4r+1}(G)$. The following lemma generalizes this observation (the case $s = 1$ implies the above observation).

LEMMA 19. *For every graph G , for every r -shallow minor H of G , and for every integer $s \geq 1$,*

$$\text{scol}_s(H) \leq \text{scol}_{2rs+2r+s}(G).$$

Proof. Let $\preceq$ be a total order of $V(G)$ such that $|R(G, u, \preceq, 2rs + 2r + s)| \leq \text{scol}_{2rs+2r+s}(G)$ for every vertex u of G . Let μ be an r -shallow model of H in G . Let ℓ_u be the leftmost vertex in $V(\mu(u))$ with respect to $\preceq$. Since the $\mu(u)$'s are pairwise disjoint, the ℓ_u are pairwise distinct. Let $\preceq'$ be the total order of $V(H)$, where $u \prec' v$ if and only if $\ell_u \prec \ell_v$.

Let $v \in V(H)$. We will show that for each $u \in R(H, v, \preceq', s)$, there exists $\tilde{u} \in V(\mu(u))$ such that $\tilde{u} \in R(G, \ell_v, \preceq, 2rs + 2r + s)$. Since the $\mu(u)$'s are pairwise disjoint, this will imply that the $\tilde{u}$'s are distinct and that $|R(H, v, \preceq', s)| \leq |R(G, \ell_v, \preceq, 2rs + 2r + s)| \leq \text{scol}_{2rs+2r+s}(G)$ for all $v \in V(H)$.

Consider a vertex $u \in R(H, v, \preceq', s)$. So $u \preceq' v$ and there is a path $(v = w_0, w_1, w_2, \dots, w_t = u)$ of length $t \leq s$ in H such that $v \prec' w_i$ for every $i \in [t - 1]$. Thus $\ell_u \preceq \ell_v \prec \ell_{w_i}$ for every $i \in [t - 1]$. For $i \in [0, t - 1]$ there is an edge $x_i y_{i+1}$ of G where $x_i \in V(\mu(w_i))$ and $y_{i+1} \in V(\mu(w_{i+1}))$. For $i \in [1, t - 1]$ there is an (y_i, x_i) -path P_i in

$\mu(w_i)$ of length at most $2r$ (since $\mu(w_i)$ has radius at most r). Similarly, there is an (ℓ_v, x_0) -path P_0 in $\mu(v)$ of length at most $2r$, and there is an (y_t, ℓ_u) -path P_t in $\mu(u)$ of length at most $2r$. Thus $P = \ell_v P_0 x_0, y_1 P_1 x_1, y_2 P_2 x_2, \dots, y_{t-1} P_{t-1} x_{t-1}, y_t P_t \ell_u$ is an (ℓ_v, ℓ_u) -path in G of length at most $(2r+1)(t+1) - 1 \leq (2r+1)(s+1) - 1 = 2rs + 2r + s$.

Let $\tilde{u}$ be the first vertex in $V(\mu(u))$ on P where $\tilde{u} \preceq \ell_v$. This is well-defined since ℓ_u is a candidate. Let P' be the subpath of P from ℓ_v to $\tilde{u}$. Consider an internal vertex x of P' . So x is in $\mu(w_i)$ for some $i \in [t]$. If $i = t$, then $\ell_v < x$ by the definition of $\tilde{u}$. If $i \in [t-1]$, then $\ell_v \prec \ell_{w_i} \preceq x$ by the definition of ℓ_{w_i} . Thus, every internal vertex of P' is to the right of ℓ_v in $\preceq$ and so $\tilde{u} \in R(G, \ell_v, \preceq, 2rs + 2r + s)$, as required. $\square$

Note that a result of Zhu [86, Corollary 3.5] implies that there is a function f that bounds the s -strong coloring numbers of an r -shallow minor H of a graph G as a function of its $f(s)$ -strong coloring number. The bound obtained in Lemma 19 is significantly better.

For weak coloring numbers, Grohe et al. [44] showed that if H is an r -shallow topological minor of G , then $\text{wcol}_s(H) \leq 2\text{wcol}_{2rs+s}(G)$ for every integer $s \geq 1$. The following lemma extends this result to shallow minors. We omit the proof since it is analogous to Lemma 19.

LEMMA 20. *For every graph G , for every r -shallow minor H of G , and for every integer $s \geq 1$,*

$$\text{wcol}_s(H) \leq \text{wcol}_{2rs+2r+s}(G).$$

For graph products, Dvořák et al. [34] showed that $\text{scol}_s(G \boxtimes H) \leq \text{scol}_s(G)(\Delta(H) + 2)^s$ for all graphs G and H . Their proof technique also applies to weak coloring numbers. A closer inspection of their proof reveals the following stronger upper bound.

LEMMA 21 (see [34]). *For all graphs G and H and every integer $s \geq 1$,*

$$\begin{aligned} \text{scol}_s(G \boxtimes H) &\leq \text{scol}_s(G)(\Delta(H^s) + 1) \quad \text{and} \\ \text{wcol}_s(G \boxtimes H) &\leq \text{wcol}_s(G)(\Delta(H^s) + 1). \end{aligned}$$

Theorem 17 follows from Lemmas 19 to 21 by setting $H = K_\ell$.

4. Shallow minors and beyond-planar graphs. This section shows that several beyond-planar graph classes can be described as a shallow minor of the strong product of a planar graph with a small complete graph. In fact, we show the slightly stronger result that they are shallow minors of the lexicographic product of a planar graph with a small edgeless graph. We then use the tools from section 3 to deduce various structural properties of these classes.

4.1. Powers of planar graphs. Recall that the k th power of a graph G is the graph G^k with vertex set $V(G^k) := V(G)$ and $uv \in E(G^k)$ if $\text{dist}_G(u, v) \leq k$ and $u \neq v$. Dujmović, Morin, and Wood [31] showed that if a graph G has maximum degree Δ , then $G^k = G^{\mathcal{P}}$ for some $(k, 2k\Delta^k)$ -shortcut system $\mathcal{P}$.

Huynh and Wood [56] introduced the following generalization of low-degree squares of graphs: for a graph G and integer $d \geq 1$, let $G^{(d)}$ be the graph obtained from G by adding a clique on $N_G(v)$ for each vertex $v \in V(G)$ with $\deg_G(v) \leq d$. Huynh and Wood [56] observed that $G^{(d)} = G^{\mathcal{P}}$ where $\mathcal{P}$ is some $(2, d)^*$ -shortcut system.

We consider a further generalization of squares of graphs. Let $d \in \mathbb{N}$ and G be a graph. For each vertex $v \in V(G)$, let $M_v \subseteq N(v)$ where $|M_v| \leq d$ and let $\mathcal{M} := \{M_v :$

$v \in V(G)\}$. Let $G^{\mathcal{M}}$ denote the graph obtained from G by adding the edge uw to G whenever there exists $v \in V(G)$ such that $u \in M_v$ and $w \in N(v)$. We call $\mathcal{M}$ a d -lift of G . Clearly $G^{(d)} \subseteq G^{\mathcal{M}}$ for some d -lift $\mathcal{M}$. So d -lifts generalize low-degree squares of graphs.

LEMMA 22. *For every graph G and every d -lift $\mathcal{M}$ of G , the graph $G^{\mathcal{M}}$ is a 1-shallow minor of $G \circ \overline{K_{d+1}}$.*

Proof. Say $\mathcal{M} = \{M_w : w \in V(G)\}$. For $u \in V(G)$, let $S_u := \{w \in V(G) : u \in M_w\} \subseteq N_G(u)$ and let $\phi_u : M_u \rightarrow \{2, \dots, d+1\}$ be an injective function. For each $u \in V(G)$, let $\mu(u)$ be the subgraph of $G \circ \overline{K_{d+1}}$ induced by $\{(u, 1)\} \cup \{(w, \phi_w(u)) : w \in S_u\}$. We claim that μ is a 1-shallow model of $G^{\mathcal{M}}$ in $G \circ \overline{K_{d+1}}$.

Let $u, w \in V(G)$ be distinct. First $\mu(u)$ is connected and has radius at most 1 since $S_u \subseteq N_G(u)$. Second, if $S_u \cap S_w = \emptyset$, then $\mu(u)$ and $\mu(w)$ are disjoint. Otherwise, there exists $v \in M_u \cap M_w$. Say $(v, i) \in \mu(u)$ and $(v, j) \in \mu(w)$. Then $i \neq j$ since ϕ_v is injective and so $\mu(u)$ and $\mu(w)$ are vertex disjoint.

It remains to show that if $uw \in E(G^{\mathcal{M}})$, then $\mu(u)$ and $\mu(w)$ are adjacent. If $uw \in E(G)$, then $\mu(u)$ and $\mu(w)$ are adjacent since $(u, 1)(w, 1) \in E(G \circ \overline{K_{d+1}})$. Otherwise, there exists $v \in V(G)$ such that $u, w \in N_G(v)$ and $u \in M_v$ or $w \in M_v$. Assume $u \in M_v$. Then $(v, \phi_v(u)) \in V(\mu(u))$ and hence $\mu(u)$ and $\mu(w)$ are adjacent since $(v, \phi_v(u))(w, 1) \in E(G \circ \overline{K_{d+1}})$, as required. $\square$

We now apply Lemma 22 to obtain the following product structure theorem for d -lifts of planar graphs.

THEOREM 23. *For every planar graph G and every d -lift $\mathcal{M}$ of G , the graph $G^{\mathcal{M}}$ is contained in $H \boxtimes P \boxtimes K_{27(d+1)}$ for some graph H with treewidth at most 19 and for some path P , and thus $\text{rtw}(G^{\mathcal{M}}) \leq 540(d+1) - 1$.*

Proof. By Theorem 2, $G \subseteq J \boxtimes P \boxtimes K_3$ for some graph J with treewidth at most 3 and some path P . By Lemma 22, $G^{\mathcal{M}}$ is a 1-shallow minor of $G \circ \overline{K_{d+1}}$. The claim then immediately follows by Theorem 10. $\square$

Since d -lifts generalize squares of graphs, we have the following corollary.

COROLLARY 24. *For every planar graph G with maximum degree Δ , the graph $G^2 \subseteq J \boxtimes P \boxtimes K_{27(\Delta+1)}$ for some graph J with treewidth at most 19, and thus $\text{rtw}(G^2) \leq 540(\Delta+1) - 1$.*

More generally, we now show that powers of graphs can be described using shallow minors.

LEMMA 25. *Let G be a graph and $k \in \mathbb{N}$ and $d := \Delta(G^{\lfloor k/2 \rfloor})$. Then G^k is a $\lfloor \frac{k}{2} \rfloor$ -shallow minor of $G \circ \overline{K_{d+1}}$.*

Proof. For a vertex $v \in V(G)$, let N_v be the set of vertices in G at distance at most $\lfloor \frac{k}{2} \rfloor$ from v . Then $|N_v| \leq d+1$, and $w \in N_v$ if and only if $v \in N_w$. Let $\phi_v : N_v \rightarrow \{1, \dots, d+1\}$ be an injective function. For each vertex $v \in V(G)$, let $\mu(v)$ be the subgraph of $G \circ \overline{K_{d+1}}$ induced by $\{(w, \phi_w(v)) : w \in N_v\}$. It follows that $\mu(v)$ is a connected subgraph of $G \circ \overline{K_{d+1}}$ with radius at most $\lfloor \frac{k}{2} \rfloor$, and $\mu(v) \cap \mu(w) = \emptyset$ for distinct $v, w \in V(G)$.

Consider an edge $uv \in E(G^k)$. Let $P_{uv} = (u = w_0, \dots, w_p = v)$ be a shortest (u, v) -path in G (and thus $p \leq k$). Then $\text{dist}_G(u, w_{\lfloor p/2 \rfloor}) \leq \lfloor \frac{k}{2} \rfloor$ and $\text{dist}_G(w_{\lfloor p/2 \rfloor + 1}, v) \leq \lfloor \frac{k}{2} \rfloor$. Thus $(w_{\lfloor p/2 \rfloor}, \phi_{w_{\lfloor p/2 \rfloor}}(u)) \in \mu(u)$ and $(w_{\lfloor p/2 \rfloor + 1}, \phi_{w_{\lfloor p/2 \rfloor + 1}}(v)) \in \mu(v)$ so

$$(w_{\lfloor p/2 \rfloor}, \phi_{w_{\lfloor p/2 \rfloor}}(u))(w_{\lfloor p/2 \rfloor + 1}, \phi_{w_{\lfloor p/2 \rfloor + 1}}(v)) \in E(G \circ \overline{K_{d+1}}).$$

Therefore μ is a $\lfloor \frac{k}{2} \rfloor$ -shallow model of G^k in $G \circ \overline{K_{d+1}}$. $\square$

Our next theorem implies that for any fixed $k \in \mathbb{N}$, if a graph G has bounded row treewidth and bounded maximum degree, then G^k will also have bounded row treewidth. Recall that $P^a \subseteq P \boxtimes K_a$ where P is a path. Applying Lemma 25 with Theorem 7 by setting $r = \lfloor \frac{k}{2} \rfloor$ and L to be path, we obtain the following.

THEOREM 26. *Let G be contained in $H \boxtimes P \boxtimes K_\ell$, where H is a graph with $\text{tw}(H) \leq t$, P is a path, and $\ell \in \mathbb{N}$. Let $k \in \mathbb{N}$ and $d := \Delta(G^{\lfloor k/2 \rfloor})$. Then $G^k \subseteq J \boxtimes P \boxtimes K_{\ell(2\lfloor k/2 \rfloor + 1)^2(d+1)}$ for some graph J with treewidth at most $\binom{2\lfloor k/2 \rfloor + 1 + t}{t} - 1$.*

In the case when G is planar, Theorem 26 with Theorem 10 (and $t = \ell = 3$) implies the following product structure for powers of G .

COROLLARY 27. *Let G be a planar graph. Let $k \in \mathbb{N}$ and $d := \Delta(G^{\lfloor k/2 \rfloor})$. Then $G^k \subseteq J \boxtimes P \boxtimes K_{3(2\lfloor k/2 \rfloor + 1)^2(d+1)}$ for some graph J with treewidth at most $\binom{2\lfloor k/2 \rfloor + 4}{3} - 1$.*

In addition, by applying Corollary 27 with Lemmas 11, 13, and 14, it follows that for every integer $k \geq 1$ and every planar graph G , the k -power G^k of G has:

- $\text{qn}(G^k) \leq (9(k+1)2^{2\lfloor k/2 \rfloor + 4})^{-1} + \frac{9}{2}(k+1)^2(\Delta(G^{\lfloor k/2 \rfloor}) + 1) \in O_k(\Delta(G^{\lfloor k/2 \rfloor}))$;
- $\pi(G^k) \leq 3(2\lfloor k/2 \rfloor + 1)2^{4\lfloor k/2 \rfloor + 4}(\Delta(G^{\lfloor k/2 \rfloor}) + 1) \in O_k(\Delta(G^{\lfloor k/2 \rfloor}))$; and
- $\chi_p(G^k) \leq 3(2\lfloor k/2 \rfloor + 1)^2(p+1)\binom{p + \binom{2\lfloor k/2 \rfloor + 4}{3} - 1}{2\lfloor k/2 \rfloor + 4 - 1}(\Delta(G^{\lfloor k/2 \rfloor}) + 1) \in O_k(p^3 \Delta(G^{\lfloor k/2 \rfloor}))$.

Observe that for all integers $k, \Delta \geq 2$, the complete $(\Delta - 1)$ -ary tree T of height $\lfloor \frac{k}{2} \rfloor$ has diameter at most k and maximum degree Δ . Since T^k is a complete graph, $\text{qn}(T^k) \geq \lfloor \frac{|V(T)|}{2} \rfloor = \lfloor \frac{\Delta(T^{\lfloor k/2 \rfloor}) + 1}{2} \rfloor$ [50] and $\pi(T^k) \geq |V(T)| = \Delta(T^{\lfloor k/2 \rfloor}) + 1$. Therefore, for fixed $k \in \mathbb{N}$, the above upper bounds $\text{qn}(G^k) \in O_k(\Delta(G^{\lfloor k/2 \rfloor}))$ and $\pi(G^k) \in O_k(\Delta(G^{\lfloor k/2 \rfloor}))$ on the queue-number and the nonrepetitive chromatic number of k -powers of planar graphs G are asymptotically best possible.

For coloring numbers of G^k , Lemma 25 and Theorem 18 together imply

$$\begin{aligned} \text{scol}_s(G^k) &\leq (\Delta(G^{\lfloor k/2 \rfloor}) + 1)(5ks + k + s + 1) \in O_k(s\Delta(G^{\lfloor k/2 \rfloor})) \quad \text{and} \\ \text{wcol}_s(G^k) &\leq (\Delta(G^{\lfloor k/2 \rfloor}) + 1)\binom{(k+1)s + k + 2}{2}((2k+2)s + 2k + 1) \in O_k(s^3\Delta(G^{\lfloor k/2 \rfloor})). \end{aligned}$$

For layered treewidth and boxicity, by applying Lemma 25 with Lemmas 15 and 16, it follows that $\text{ltw}(G^k) \leq 3(\Delta(G^{\lfloor k/2 \rfloor}) + 1)(2k + 1)$ and $\text{box}(G^k) \leq 18(\Delta(G^{\lfloor k/2 \rfloor}) + 1)(2k + 1) + 3$.

4.2. (g, k) -planar graphs. Recall that a graph G is k -planar if it is isomorphic to an embedded graph G' in the plane where each edge in G' is involved in at most k crossings. This definition has a natural extension for other surfaces. A graph G' embedded on a surface Σ is (Σ, k) -plane if every edge of G' is involved in at most k crossings. A graph is (g, k) -planar if it is isomorphic to some (Σ, k) -plane graph for some surface Σ with Euler genus at most g .

Dujmović, Morin, and Wood [31] observed that every (g, k) -planar graph is a subgraph of $G^{\mathcal{P}}$ for some graph G with Euler genus at most g and some $(k + 1, 2)$ -shortcut system $\mathcal{P}$. Thus, by Lemma 6, every (g, k) -planar graph is a $\frac{k}{2}$ -shallow topological minor of $G \circ \overline{K_3}$ where G has Euler genus at most g . We obtain a slightly stronger bound by the standard planarization method.

LEMMA 28. *Every (g, k) -planar graph G is a $\frac{k}{2}$ -shallow topological minor of $H \circ \overline{K_2}$ where H has Euler genus at most g .*

Proof. Embed G into a surface Σ with Euler genus at most g such that every edge of G is involved in at most k crossings. For each crossing $(p, \{uv, xy\})$ where $uv, xy \in E(G)$, insert a dummy vertex w at p to obtain a graph H with Euler genus at most g . Let $W := V(H) \setminus V(G)$ be the set of dummy vertices. Each $uv \in E(G)$ corresponds to a path P_{uv} with length at most k in H where each internal vertex of P is a dummy vertex. Let $\mathcal{P}$ be the set of such paths. For each $w \in W$, let M_w be the set of paths in $\mathcal{P}$ that contains w as an internal vertex. Then $|M_w| \leq 2$. Let $\phi_w : M_w \rightarrow \{1, 2\}$ be an injective function. For each $v \in V(G)$, let $\phi_v(P) := 1$ for all paths P_{uv} with end-vertex v . For each path $P_{uv} = (u = w_0, w_1, \dots, w_{j-1}, w_j = v) \in \mathcal{P}$ where $j \leq k+1$, let $\tilde{P}_{uv}$ be the path in $H \circ \overline{K_2}$ defined by $V(\tilde{P}_{uv}) := \{(w_i, \phi_{w_i}(P_{uv})) : i \in [0, j]\}$ and $E(\tilde{P}_{uv}) := \{(w_i, \phi_{w_i}(P_{uv}))(w_{i+1}, \phi_{w_{i+1}}(P_{uv})) : i \in [0, j-1]\}$. Let $\tilde{\mathcal{P}}$ be the set of such $\tilde{P}_{uv}$ paths.

We claim that $\tilde{\mathcal{P}}$ defines a $\frac{k}{2}$ -shallow topological minor of G in $H \circ \overline{K_2}$ where each vertex $v \in V(G)$ is mapped to $(v, 1) \in V(H \circ \overline{K_2})$ and each edge $uv \in E(G)$ is mapped to $\tilde{P}_{uv} \in \tilde{\mathcal{P}}$.

Let $uv \in E(G)$. Then there is a path $P_{uv} \in \mathcal{P}$ with length at most $k+1$ and end-vertices u and v . By construction, $\tilde{P}_{uv}$ is a path in $H \circ \overline{K_2}$ from $(u, 1)$ to $(v, 1)$ with length at most k . Thus, it suffices to show that the paths in $\tilde{\mathcal{P}}$ are internally disjoint. Now for a path $\tilde{P}_{uv} \in \tilde{\mathcal{P}}$, its internal vertices are of the form $(w, \phi_w(P_{uv}))$ where $w \in W$. Now if there is another path $\tilde{Q}_{xy} \in \tilde{\mathcal{P}}$ for which $(w, \phi_w(P_{uv}))$ is an internal vertex of $\tilde{Q}_{xy}$, then $\tilde{P}_{uv} = \tilde{Q}_{xy}$ since ϕ_w is injective. As such, the paths in $\tilde{\mathcal{P}}$ are internally disjoint, as required. $\square$

Applying Lemma 28 with Theorems 3 and 10, we obtain the following product structure theorems for (g, k) -planar graphs.

THEOREM 29. *Every (g, k) -planar graph G is contained in $H \boxtimes P \boxtimes K_{2 \max\{2g, 3\}(k+1)^2}$ for some graph H with treewidth at most $\binom{k+4}{3} - 1$, and thus G has row treewidth at most $2 \max\{2g, 3\}(k+1)^2 \binom{k+4}{3} - 1$.*

Note that Dujmović, Morin, and Wood [31] proved that every (g, k) -planar graph is a subgraph of $H \boxtimes P \boxtimes K_{\max\{2g, 3\}(6k^2+16k+10)}$ for some graph H of treewidth at most $\binom{k+4}{3} - 1$. Thus, our results only improve those of Dujmović, Morin, and Wood [31] by a constant factor. We do not state the bounds that we obtain for the queue-number, nonrepetitive chromatic number, and p -centered chromatic number as the improvement is not substantial.

For coloring numbers, applying Lemma 28 with Theorem 18, it follows that for every k -planar graph G ,

$$\begin{aligned} \text{scol}_s(G) &\leq 10(k+1)s + 20k + 2 \quad \text{and} \\ \text{wcol}_s(G) &\leq \binom{(k+1)s+k+2}{2} (4(k+1)s + 4k + 2), \end{aligned}$$

and for every (g, k) -planar graph G ,

$$\begin{aligned} \text{scol}_s(G) &\leq (8g+10)(k+1)s + (8g+10)k + 4g + 2 \quad \text{and} \\ \text{wcol}_s(G) &\leq (2g + \binom{(k+1)s+k+2}{2})((2k+4)s + 4k + 2). \end{aligned}$$

These results for strong coloring numbers differ from the previous bounds by van den Heuvel and Wood [82] (obtained via layered treewidth) by a small constant factor. More importantly, these are the first known polynomial bound on weak coloring numbers for k -planar and (g, k) -planar. Superpolynomial bounds were previously known for (g, k) -planar graphs since they have bounded strong coloring numbers [86].

4.3. String graphs. A *string graph* is the intersection graph of a set of curves in the plane with no three curves meeting at a single point. For an integer $\delta \geq 2$, if each curve is involved in at most δ intersections with other curves, then the corresponding string graph is called a δ -*string graph*. A (g, δ) -*string graph* is defined analogously for curves on a surface with Euler genus at most g .

Dujmović, Morin, and Wood [31] showed that every (g, δ) -string graph G is a subgraph of $H^{\mathcal{P}}$ for some graph H with Euler genus at most g and some $(\delta+1, \delta+1)$ -shortcut system $\mathcal{P}$. By Lemma 6, G is a $\frac{\delta}{2}$ -shallow topological minor of $H \circ \overline{K}_{\delta+2}$. We obtain a stronger bound by the standard planarization method.

LEMMA 30. *Every (g, δ) -string graph G is a $\lfloor \frac{\delta}{2} \rfloor$ -shallow minor of $H \circ \overline{K}_2$ for some graph H with Euler genus at most g .*

Proof. Let $\mathcal{C} := \{C_v : v \in V(G)\}$ be a set of curves in a surface of Euler genus at most g such that each curve is involved in at most δ intersections with other curves and the intersection graph is isomorphic to G . For each pair of curves in $\mathcal{C}$ that intersects, add a vertex at their intersection point. Add an edge between each pair of consecutive vertices along a curve in $\mathcal{C}$. If a curve C_v is involved in exactly one crossing, add another vertex c_v to the curve adjacent to the intersection point that C_v contains. If a curve C_v is not involved in any crossing, add a vertex c_v to the curve. Let H be the planar graph obtained by this process. For each vertex $v \in V(G)$, let M_v be the set of vertices in H that are on the curve C_v . For each $w \in V(H)$, let $N_w := \{v \in V(G) : w \in M_v\}$. Then $|N_w| \leq 2$ for all $w \in V(H)$.

Observe that $H[M_v]$ has radius at most $\lfloor \frac{\delta}{2} \rfloor$ for each vertex $v \in V(G)$. For each $w \in V(H)$, let $\phi_w : N_w \rightarrow \{1, 2\}$ be an injective function. For each vertex $v \in V(G)$, let $\mu(v)$ be the subgraph of $H \circ \overline{K}_2$ induced by $\{(w, \phi_w(v)) : w \in M_v\}$.

We claim that μ defines a $\lfloor \frac{\delta}{2} \rfloor$ -shallow model of G in $H \circ \overline{K}_2$. Let $u, v \in V(G)$ be distinct. First, since $H[M_v]$ has radius at most $\lfloor \frac{\delta}{2} \rfloor$ and is connected, so is $\mu(u)$. Second, if u and v do not intersect, then $\mu(u)$ and $\mu(v)$ are disjoint. Otherwise, there exists $w \in C_u \cap C_v$, in which case $\phi_w(u) \neq \phi_w(v)$ since ϕ_w is injective and hence $\mu(u)$ and $\mu(v)$ are vertex disjoint. Finally, if $uv \in E(G)$, then there is a vertex $w_1 \in M_u \cap M_v$. Let $w_2 \in M_v$ be a neighbor of w_1 . Then $(w_1, \phi_{w_1}(u))(w_2, \phi_{w_2}(v)) \in E(H \circ \overline{K}_2)$ so $\mu(u)$ and $\mu(v)$ are adjacent. Therefore μ defines a $\lfloor \frac{\delta}{2} \rfloor$ -shallow model of G in $H \circ \overline{K}_2$. $\square$

Applying Lemma 30 in conjunction with Theorems 3 and 10, we deduce the following product structure theorem for string graphs.

THEOREM 31. *Every (g, δ) -string graph G is contained in $H \boxtimes P \boxtimes K_{2 \max\{2g, 3\}(\delta+1)^2}$ for some graph H with treewidth at most $\binom{2\lfloor \delta/2 \rfloor + 4}{3} - 1$, and thus G has row treewidth at most $2 \max\{2g, 3\}(\delta+1)^2 \binom{2\lfloor \delta/2 \rfloor + 4}{3} - 1$.*

Note that Dujmović, Morin, and Wood [31] previously showed that every (g, δ) -string graph is contained in $H \boxtimes P \boxtimes K_{\max\{2g, 3\}(\delta^4 + 4\delta^3 + 9\delta^2 + 10\delta + 4)}$ for some graph H with treewidth at most $\binom{\delta+4}{3} - 1$. Thus, Theorem 31 improves this result by a factor of δ^2 in the K_ℓ term.

For coloring numbers, by applying Lemma 30 with Theorem 18, it follows that for every (g, δ) -string graph G ,

$$\begin{aligned} \text{scol}_s(G) &\leq (8g + 10)(\delta + 1)s + (8g + 10)\delta + 4g + 2 \quad \text{and} \\ \text{wcol}_s(G) &\leq 2(2g + \binom{(\delta+1)s + \delta + 2}{2})((2\delta + 2)s + 2\delta + 1). \end{aligned}$$

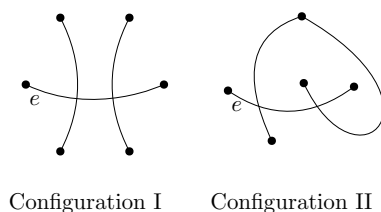


FIG. 3. Forbidden crossing configurations for fan-planar graphs.

4.4. Fan-planar graphs. Recall that an embedded graph G is fan-planar if for each edge $e \in E(G)$, the edges that cross e have a common end-vertex and they cross e from the same side (when directed away from their common end-vertex). Equivalently, an embedded graph is fan-planar if it forbids the two crossing configurations in Figure 3.³ Fan-planar graphs were introduced by Kaufmann and Ueckerdt [58]. The class of fan-planar graphs extends 1-planar graphs and is a proper subset of 3-quasi planar graphs.

We now work toward showing that fan-planar graphs have a shallow-minor structure. Note that compared to the other graph classes that we consider in this paper, the proof for this result is much more involved and is highly nontrivial.

For an embedded fan-planar graph G and a crossed edge $uv \in E(G)$, a common end-vertex of the edges that cross uv is called a *friend* of uv . If uv is crossed at least twice, then it has one friend; otherwise, uv is crossed once and has two friends. A *friend assignment* of G assigns a friend to each crossed edge. For a given friend assignment, a crossed edge uv is *well-behaved* if there exists a noncrossing point $p \in uv$ such that u is the assigned friend of each edge that crosses uv between p and u , and v is the assigned friend of each edge that crosses uv between p and v . If every crossed edge in G is well-behaved, then the friend assignment is *well-behaved*.

LEMMA 32. *Every simple fan-planar graph G has a well-behaved friend assignment.*

Proof. Let $uv \in E(G)$ be a crossed edge and let $e_1, \dots, e_m$ be the edges that cross uv ordered by their crossing point from u to v . Let w be a common end-vertex of $e_1, \dots, e_m$. If none of $e_1, \dots, e_m$ cross another edge incident to v , then let u be the assigned friend of $e_1, \dots, e_m$ and choose p to be an arbitrary point (along uv) between $e_m \cap uv$ and v . Otherwise, let $i \in [m]$ be minimum such that e_i crosses another edge vz incident to v . Choose p to be an arbitrary point on uv between $uv \cap e_{i-1}$ and $uv \cap e_i$. Since none of the edges $e_1, \dots, e_{i-1}$ cross another edge incident to v , we may let u be their assigned friend.

It remains to show that for each $j \in [i, m]$, v is a common end-vertex of the edges that cross e_j and thus we may let v be the assigned friend of e_j . If e_j is only crossed by uv , then we are done. If e_j crosses vz , then v is e_j 's only friend. Otherwise, the end-vertex of e_j opposite to w is contained in the region bounded by the edges uv, vz, e_i (since G is simple fan-planar). Now suppose that another edge $\tilde{e}$ incident to u crosses e_j . Since $\tilde{e}$ does not cross e_i or vz (as Configuration I is forbidden), $\tilde{e}$ must cross e_j from the opposite side that uv cross e_j , contradicting G being fan-planar.

³Note that there is another notion of *fan-planarity* in the literature which includes an additional excluded crossing configuration (see [17, 19]). Our results hold for both understandings of fan-planarity.

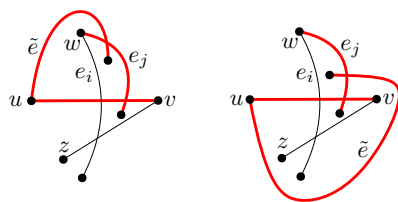
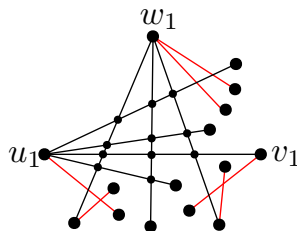


FIG. 4. Proof of Lemma 32.

FIG. 5. The graph $G'_1 := \tilde{G}_1 \cup (G^{(0)} - E_1)$ where black edges are from $E(\tilde{G}_1)$ and red edges are from $E(G^{(0)}) - E_1$.

(see Figure 4). Thus uv is the only edge incident to u that crosses e_j and so v is a common end-vertex of the edges that cross e_j , as required. $\square$

We now prove our main technical lemma of this subsection.

LEMMA 33. *Every fan-planar graph G is a 1-shallow minor of $H \circ \overline{K_3}$ for some planar graph H .*

Proof. By [61, Theorem 1] and Lemma 32, we may assume that G has a simple fan-planar embedding with a well-behaved friend assignment. Initialize $G^{(0)} := G$. Arbitrarily choose an edge $u_1v_1 \in E(G^{(0)})$ that is involved in at least two crossings and let $w_1 \in V(G^{(0)})$ be the assigned friend of u_1v_1 . If no such edge exists, then G is 1-planar and we are done by Lemma 28. Let E_1 be a maximal subset of $E(G^{(0)})$ such that:

- $u_1v_1 \in E_1$;
- each edge in E_1 has u_1 , v_1 , or w_1 as its assigned friend; and
- $E_1 \setminus V(G^{(0)})$ is a connected subset of the plane.

Observe that every edge in E_1 crosses another edge in E_1 and thus every edge in E_1 is incident to u_1, v_1 , or w_1 .

Let V_1 be the set of vertices incident to edges in E_1 . Consider the subgraph of $G^{(0)}$ with vertex set V_1 and edge set E_1 . Add a dummy vertex at each crossing point to planarize the subgraph. Let $\tilde{G}_1$ be the plane graph obtained and let $\tilde{D}_1$ be the set of dummy vertices added. Let $G'_1 := \tilde{G}_1 \cup (G^{(0)} - E_1)$; see Figure 5.

Claim 1. If an edge $e \in E(\tilde{G}_1)$ is crossed in G'_1 , then e is incident to some $z \in V_1 \setminus \{u_1, v_1, w_1\}$.

Proof. Since $\tilde{G}_1$ is plane, e is crossed by some edge $e' \in E(G^{(0)}) \setminus E_1$. Now e is a segment of some edge $xz \in E_1$ where $x \in \{u_1, v_1, w_1\}$ and $z \in V_1$. For the sake of contradiction, suppose that e is not incident to any vertex in $V_1 \setminus \{u_1, v_1, w_1\}$. Then there exist an edge $\tilde{e} \in E_1$ and a dummy vertex $d = (xz \cap \tilde{e}) \in \tilde{D}_1$ such that e is between x and d (along xz). Now if the assigned friend of $\tilde{e}$ is x , then the assigned friend of e' is also x since xz is well-behaved. Otherwise, if the assigned friend of $\tilde{e}$ is z ,

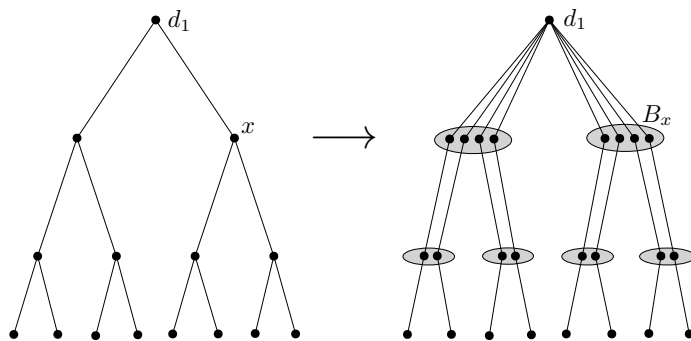
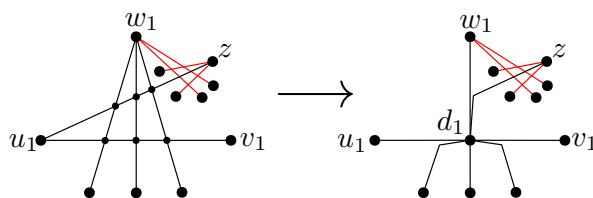


FIG. 6. Replacing an embedded tree by a subdivided star.

FIG. 7. Obtaining $G^{(1)}$ from G'_1 .

then $z \in \{u_1, v_1, w_1\} \setminus \{x\}$. Thus, the assigned friend of e' is in $\{x, z\} \subseteq \{u_1, v_1, w_1\}$. However, this contradicts the maximality of E_1 since e' crosses an edge in E_1 and the assigned friend of e' is in $\{u_1, v_1, w_1\}$, as required. $\square$

For $\epsilon > \delta > 0$, for every vertex $x \in V(G'_1)$ and edge $uv \in E(G'_1)$, let

$$B_x := \{p \in \mathbb{R}^2 : \text{dist}_{\mathbb{R}^2}(p, x) \leq \epsilon\} \quad \text{and} \quad C_{uv} := \{p \in \mathbb{R}^2 : \text{dist}_{\mathbb{R}^2}(p, uv) \leq \delta\} \setminus (B_u \cup B_v).$$

Choosing ϵ and δ to be sufficiently small, we may assume that $B_x \cap B_y = \emptyset$, $B_x \cap C_{uv} = \emptyset$, and $C_{uv} \cap C_{ab} = \emptyset$ for all $x, y \in V(G'_1)$ and pairwise noncrossing edges $ab, uv \in E(G'_1)$. Let $T^{(1)}$ be a spanning tree of $\tilde{G}_1$ rooted at some $d_1 \in \tilde{D}_1$. Using a standard blow-up trick (e.g., see [56, Lemma 5.5]), we may replace $T^{(1)}$ by a subdivided star $S^{(1)}$ embedded in the region $(\bigcup_{x \in V(\tilde{G}_1)} B_x) \cup (\bigcup_{uv \in E(\tilde{G}_1)} C_{uv})$, where each root to leaf path in $S^{(1)}$ corresponds to a root to leaf path in $T^{(1)}$; see Figure 6.

Let $G^{(1)}$ be the embedded graph obtained from G'_1 by replacing $\tilde{G}_1$ by $S^{(1)}$ then removing the subdivision vertices in $S^{(1)}$; see Figure 7. Charge u_1, v_1 , and w_1 to d_1 . By Claim 1, for every crossed edge of the form $d_1 z \in E(G^{(1)})$ where $z \in V_1$, there exists an edge $xz \in E_1$ where $x \in \{u_1, v_1, w_1\}$ such that every edge that crosses $d_1 z$ (in $G^{(1)}$) also crosses xz (in $G^{(0)}$) in the same direction.

Claim 2. $G^{(1)}$ is simple fan-planar and there is a well-behaved friend assignment of $G^{(1)}$ where d_1 is not an assigned friend of any edge.

Proof. We first show that $G^{(1)}$ is simple fan-planar. Let $e \in E(G^{(1)})$ be a crossed edge and let A_e be the set of edges in $E(G^{(1)})$ that cross e . If $\{e\} \cup A_e \subseteq E(G^{(0)})$, then Configuration I and Configuration II are forbidden and e does not cross any edge at least twice since $G^{(0)}$ is simple fan-planar. So assume that $(\{e\} \cup A_e) \cap (E(G^{(1)}) \setminus E(G^{(0)})) \neq \emptyset$. There are two cases to consider.

First, suppose $e = d_1 z$ for some $z \in V_1$. Since the edges in $E(G^{(1)}) \setminus E(G^{(0)})$ pairwise do not cross, it follows that $A_e \subseteq E(G^{(0)})$ and e does not cross any edge

incident to d_1 . By Claim 1, there is an edge $xz \in E_1$ where $x \in \{u_1, v_1, w_1\}$ such that every edge that crosses e (in $G^{(1)}$) also crosses xz (in $G^{(0)}$) in the same direction. Thus the edges that cross e have a common end-vertex and they all cross e in the same direction and e does not cross any edge more than once or any edge incident to z .

Second, suppose $e \in E(G^{(1)}) \cap E(G^{(0)})$ and $A_e \cap (E(G^{(1)}) \setminus E(G^{(0)})) \neq \emptyset$. Let y be the assigned friend of e in $G^{(0)}$. Since e crosses an edge in $E(G^{(1)}) \setminus E(G^{(0)})$ and E_1 is maximal, it follows that $y \in V_1 \setminus \{u_1, v_1, w_1\}$. Thus $A_e \cap (E(G^{(1)}) \setminus E(G^{(0)})) = \{d_1y\}$ since $y \notin \{u_1, v_1, w_1\}$. Furthermore, every edge in $A_e \cap E(G^{(0)})$ is incident to y since y is the friend of e in $G^{(0)}$. As such, every edge that crosses e (in $G^{(1)}$) is incident to y . By Claim 1, there is an edge $xy \in E_1$ where $x \in \{u_1, v_1, w_1\}$ such that every edge that crosses d_1y (in $G^{(1)}$) crosses xy (in $G^{(0)}$) in the same direction. Thus Configurations I and II are forbidden and e does not cross any edge more than once since $G^{(0)}$ is simple fan-planar, as required.

We now specify the friend assignment. By Lemma 32, G has a well-behaved friend assignment since it is simple fan-planar. Now suppose d_1 is the assigned friend of some edge e . Since the edges u_1d_1, v_1d_1 , and w_1d_1 are uncrossed in $G^{(1)}$, e crosses an edge of the form d_1y where $y \in V_1 \setminus \{u_1, v_1, w_1\}$. By an analogous argument to that in the previous paragraph, every edge that crosses e in $G^{(1)}$ is also incident to y , we may modify the friend assignment so that y is the assigned friend of e . By doing so for every such edge, we obtain the desired friend assignment. $\square$

Observe the following properties of $G^{(1)}$:

- edges that are uncrossed in $G^{(0)}$ remain uncrossed in $G^{(1)}$;
- the edges u_1d_1, v_1d_1 , and w_1d_1 are uncrossed in $G^{(1)}$; and
- $G^{(1)}$ has fewer crossings than $G^{(0)}$.

Let $(u_1, v_1, w_1, E_1, d_1, G^{(1)}), (u_2, v_2, w_2, E_2, d_2, G^{(2)}), \dots, (u_m, v_m, w_m, E_m, d_m, G^{(m)})$ be the sequence obtained by iterating the above procedure where $u_iv_i \in E(G^{(i-1)})$ is an arbitrary edge crossed at least twice in $G^{(i-1)}$, d_i is not an assigned friend of any edge in $G^{(j)}$ for all $i \leq j$, and each edge in $G^{(m)}$ is crossed at most once. Then $G^{(m)}$ is 1-planar. Note that this sequence is well-defined since $G^{(i)}$ has less crossings than $G^{(i-1)}$.

Add a dummy vertex at each crossing point in $G^{(m)}$ and let H be the plane graph obtained. For every edge of the form $d_iz \in E(G^{(m)})$ that is crossed by another edge e , charge z as well as the end-vertices of e to the dummy vertex at $d_iz \cap e$. For crossed edges of the form $e', \tilde{e} \in E(G) \cap E(G^{(m)})$, charge an end-vertex of e' and an end-vertex of $\tilde{e}$ to the dummy vertex at $e' \cap \tilde{e}$.

Let $D := V(H) \setminus V(G)$ be the set of dummy vertices in H . Observe that every dummy vertex $d_i \in D$ has at most three vertices charged to it, u_i, v_i, w_i , and that these vertices are neighbors of d_i in H . In addition, the original edges in G have the following key properties.

Claim 3. If $xy \in E(G)$, then either:

Case 1. $xy \in E(H)$;

Case 2. there is a path $P_{xy} = (x, d, y)$ in H where x or y is charged to d and $d \in D$; or

Case 3. there is a path $P_{xy} = (x, d_1, d_2, y)$ in H where x is charged to d_1 and y is charged to d_2 and $d_1, d_2 \in D$.

Proof. First, consider an edge $xy \in E(G^{(m)}) \cap E(G)$. If xy is uncrossed in $G^{(m)}$, then $xy \in E(H)$ and we are in Case 1. Otherwise xy is involve in a single crossing in $G^{(m)}$ and so we are in Case 2. So assume $xy \in E_i$ for some $i \in [m]$. Since every edge in E_i is incident to u_i, v_i , or w_i , we may assume that $x \in \{u_i, v_i, w_i\}$. Thus x is charged

to d_i and $d_i x \in E(H)$. If $d_i y \in E(G^{(m)})$ and it is uncrossed, then we are in Case 2. If $d_i y \in E(G^{(m)})$ and it is crossed, then we are in Case 3. So it remains to consider the case when $d_i y \in E_j$ for some $j \in [i+1, m]$. Since d_i is not the assigned friend of any edge in $G^{(j)}$, it follows that $y \in \{u_j, v_j, w_j\}$. Thus every edge e that crosses $d_i y$ is contained in E_j as y is the assigned friend of e . As such, after planarizing E_j (by replacing it with a star rooted at d_j), y will be charged to d_j and the edges $d_i d_j$ and $d_j y$ will be uncrossed in $G^{(j)}$, and hence we are in Case 3. $\square$

To finish the proof, we now specify the model μ for the 1-shallow minor of G in $H \circ \overline{K_3}$. For each dummy vertex $d \in D$, let $\phi_d: N_d \rightarrow \{1, 2, 3\}$ be an injective function where N_d is the set of vertices that are charged to d . For a vertex $u \in V(G)$, let M_u be the set of dummy vertices that u is charged to. For each $u \in V(G)$, let $\mu(u)$ be the subgraph of $H \circ \overline{K_3}$ induced by $\{(u, 1)\} \cup \{(d, \phi_d(u)) : d \in M_u\}$.

Let $u, v \in V(G)$ be distinct. First $\mu(u)$ is connected and has radius at most 1 since $ud \in E(H)$ for all $d \in M_u$. Second, if $M_u \cap M_v = \emptyset$, then clearly $\mu(u)$ and $\mu(v)$ are disjoint. Otherwise, if there exists some $d \in M_u \cap M_v$, then $\phi_d(u) \neq \phi_d(v)$ since ϕ_d is injective. Thus, $\mu(u)$ and $\mu(v)$ are vertex disjoint. Finally, Claim 3 implies that if $uv \in E(G)$, then $\mu(u)$ and $\mu(v)$ are adjacent. Therefore μ defines a 1-shallow model of G in $H \circ \overline{K_3}$. $\square$

Applying Lemma 33 with Theorems 2 and 10, we obtain the following product structure theorem for fan-planar graphs.

THEOREM 34. *Every fan-planar graph G is contained in $H \boxtimes P \boxtimes K_{81}$ for some graph H with treewidth at most 19, and thus G has row treewidth at most 1619.*

In addition, by applying Theorem 34 with Lemmas 11, 13, and 14, it follows that for every fan-planar graph G ,

- $\text{qn}(G) \leq 3 \times 81 \times 2^{19} + 3 \times \lfloor 81/2 \rfloor = 127402104$;
- $\pi(G) \leq 81 \times 4^{20}$; and
- $\chi_p(G) \leq 81(p+1) \binom{p+19}{19} \in O(p^{20})$.

For coloring numbers, by applying Lemma 33 in conjunction with Theorem 18, it follows that for every fan-planar graph G ,

$$\text{scol}_s(G) \leq 45s + 33 \quad \text{and} \quad \text{wcol}_s(G) \leq \binom{3s+4}{2} (18s + 15).$$

For layered treewidth and boxicity, by applying Lemma 33 with Lemmas 15 and 16, it follows that $\text{ltw}(G) \leq 45$ and $\text{box}(G) \leq 276$ for every fan-planar graph G .

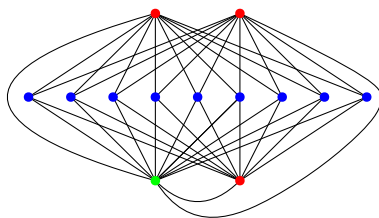
We now show that fan-planar graphs cannot be described by applying a shortcut system to a planar graph.

Binucci et al. [7, Theorem 3] proved that for every integer $k \geq 1$, there is a fan-planar graph that is not k -planar. Essentially the same proof gives the following, stronger result.

PROPOSITION 35. *For every integer $k \geq 1$, there is a fan-planar graph that is not k -gap planar.*

Proof. For each $h \in \mathbb{N}$, the complete tripartite graph $K_{1,3,h}$ is fan-planar and $|E((K_{1,3,h}))| = 4h + 3$; see Figure 8. Moreover, it is known [4, Theorem 1] that $\text{cr}(K_{1,3,h}) = \Omega(h^2) = \Omega(|E((K_{1,3,h}))|^2)$. Thus, this graph family has superlinear crossing number so it is not k -gap planar. $\square$

Lemma 5 and Proposition 35 imply that fan-planar graphs cannot be described by shortcut systems applied to a planar graph. This highlights the value and power of shallow minors.

FIG. 8. A fan-planar embedding of $K_{1,3,h}$.

As a converse to Proposition 35, Binucci et al. [7, Theorem 4] showed that there exists a 2-planar graph that is not fan-planar. Since every 2-planar graph is a 1-gap planar graph [5, Theorem 7], we have the following.

PROPOSITION 36. *For every integer $k \geq 1$, there is a k -gap planar graph that is not fan-planar.*

Propositions 35 and 36 demonstrate that for any integer $k \geq 1$, the class of fan-planar graphs and k -gap planar graphs are incomparable.⁴ This resolves an open problem of Didimo, Liotta, and Montecchiani [22, Problem 12], who asked for the relationship between k -gap planar graphs and fan-planar graphs.

4.5. k -fan-bundle planar graph. In a k -fan-bundle embedding of a graph in the plane the edges of a fan may be bundled together at their end-vertices and crossings between bundles are allowed as long as each bundle is crossed by at most k other bundles. More formally, in a k -fan-bundle planar embedding of a graph G , each edge has three parts; the first and the last parts are *fan-bundles*, which may be shared by several edges in a fan, while the middle part is unbundled. Each fan-bundle can cross at most k other fan-bundles, while the unbundled parts are crossing-free. A vertex $u \in V(G)$ can be incident to more than one bundle. Let B_u be one such bundle. We say that B_u is *anchored* at u which is the *origin* of B_u . The endpoint of B_u that is different from u is the *terminal* of B_u . A graph is *k -fan-bundle planar* if it admits a *k -fan-bundle-planar embedding*. Fan bundle planar graphs were introduced by Angelini et al. in [3], where they studied their density and algorithmic properties. Our results are the first to consider these graphs from a structural perspective.

LEMMA 37. *Every k -fan-bundle planar graph G is a $(k+1)$ -shallow minor of $H \circ \overline{K_2}$ for some planar graph H .*

Proof. Begin with a k -fan-bundle-planar embedding of G . For each bundle $B_u^{(i)}$, add a dummy vertex at the terminal of $B_u^{(i)}$ to obtain a k -planar graph H' . Let W be the set of dummy vertices added. Replace each crossing in H' by a dummy vertex to obtain a plane graph H (see Figure 9) and let D be the new dummy vertices that are added at this step. In doing so, each bundle $B_u^{(i)}$ is replaced by a (u, w) -path $P_u^{(i)}$ on at most $k+2$ vertices for some dummy vertex $w \in W$. For each $d \in D$, let $\phi_d : \{u, v\} \rightarrow \{1, 2\}$ be an injective function where u and v are the origins of the bundles that cross at d . For each $u \in V(G)$, let $W_u := (\bigcup_{i \in [c_u]} V(P_u^{(i)})) \cap W$ and $D_u := (\bigcup_{i \in [c_u]} V(P_u^{(i)})) \cap D$ where c_u is the number of bundles anchored at u . For each $u \in V(G)$, let $\mu(u)$ be the subgraph of $H \circ \overline{K_2}$ induced by $\{(u, 1)\} \cup \{(w, 1) : w \in W_u\} \cup \{(d, \phi_w(d)) : d \in D_u\}$.

⁴Graph classes $\mathcal{G}_1$ and $\mathcal{G}_2$ are *incomparable* if $\mathcal{G}_1 \setminus \mathcal{G}_2 \neq \emptyset$ and $\mathcal{G}_2 \setminus \mathcal{G}_1 \neq \emptyset$.

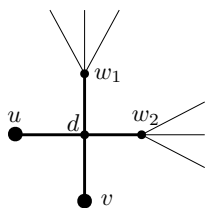


FIG. 9. Planarizing a 1-fan-bundle planar graph.

We claim that μ is a $(k+1)$ -shallow model of G in $H \circ \overline{K_2}$. Let $u, v \in V(G)$ be distinct. First, $\mu(u)$ is connected with radius at most $k+1$ as it is the union of paths on at most $k+2$ vertices that share a common end-vertex. Now if no bundle anchored at u crosses a bundle anchored at v , then $\mu(u)$ and $\mu(v)$ are clearly disjoint. Otherwise, if there is a bundle $B_u^{(i)}$ that crosses a bundle $B_v^{(i')}$, then there is some $d \in D_u \cap D_v$, in which case, $\phi_d(u) \neq \phi_d(v)$ since ϕ_d is injective and so $\mu(u)$ and $\mu(v)$ are vertex disjoint. Now if $uv \in E(G)$, then there is a bundle $B_u^{(i)}$ anchored at u and a bundle $B_v^{(i')}$ anchored at v that are adjacent. Let $w_1, w_2 \in W$ be the dummy vertices that are respectively added to the terminals of $B_u^{(i)}$ and $B_v^{(i')}$. Then $(w_1, 1) \in \mu(u)$ and $(w_2, 1) \in \mu(v)$ so $(w_1, 1)(w_2, 1) \in E(H \circ \overline{K_2})$ and thus $\mu(u)$ and $\mu(v)$ are adjacent, as required.

Applying Lemma 37 with Theorems 2 and 10, we obtain the following product structure theorem for k -fan-bundle planar graphs.

THEOREM 38. *Every k -fan-bundle planar graph G is contained in $H \boxtimes P \boxtimes K_{6(2k+3)^2}$ for some graph H with treewidth at most $\binom{2k+6}{3} - 1$, and thus G has row treewidth at most $\binom{2k+6}{3} 6(2k+3)^2 - 1$.*

Applying Theorem 38 with Lemmas 11, 13, and 14, it follows that every k -fan-bundle planar graph G has:

- $\text{qn}(G) \leq 18(2k+3)^2 2^{\binom{2k+6}{3}-1} + 9(2k+3)^2 \in 2^{O(k^3)}$;
- $\pi(G) \leq 6(2k+3)^2 4^{\binom{2k+6}{3}} \in 2^{O(k^3)}$; and
- $\chi_p(G) \leq 6(2k+3)^2 (p+1) \binom{p+\binom{2k+6}{3}-1}{\binom{2k+6}{3}-1} \in p^{O(k^3)}$.

Note that by using the shallow-minor structure of k -fan-bundle planar graph (Lemma 37) together with Lemma 12 and $(\star)$, and the fact that planar graphs have queue-number at most 42 [6], it follows that for every k -fan-bundle planar graph G ,

$$\text{qn}(G) \leq (2k+3)(2 \times ((2 \times 2 - 1)42 + 2 - 1))^{2k+3} = (2k+3)(254)^{2k+3} \in 2^{O(k)}.$$

Thus we obtain a stronger bound for the queue-number of k -fan-bundle planar graph using their shallow-minor structure instead of their product structure.

For coloring numbers, Lemma 37 and Theorem 18 imply that for every k -fan-bundle planar graph G ,

$$\begin{aligned} \text{scol}_s(G) &\leq (20k+30)s + 20k + 22 \quad \text{and} \\ \text{wcol}_s(G) &\leq \binom{(2k+3)s+2k+4}{2} ((8(k+3)s + 8k + 10)). \end{aligned}$$

For layered treewidth and boxicity, it follows from Lemmas 15, 16, and 37, that $\text{ltw}(G) \leq 24k + 25$ and $\text{box}(G) \leq 144k + 154$ for every k -fan-bundle planar graph G .

We raise the following open problems concerning k -fan-bundle graphs. Angelini et al. [3] showed that 1-fan-bundle graphs are incomparable with 2-planar graphs.

What is the relationship between 1-fan-bundle graphs and k -planar graphs? Does there exist an integer $k \geq 1$ such that every 1-fan-bundle graph is k -planar? Or more weakly, does there exist an integer $k \geq 1$ such that every 1-fan-bundle graph is k -gap planar?

5. Lower bounds. Having described several beyond-planar graph classes as shallow minors of the strong product of a planar graph with a small complete graph, we now give examples of classes that cannot be described in this manner.

5.1. k -gap planar. Recall that a graph G is k -gap planar if it is isomorphic to an embedded graph where each crossing is charged to one of the two edges involved and each edge has at most k crossings charged to it. This class of graphs has been implicitly studied for some time [39, 71]. The language of k -gap planar graphs was introduced by Bae et al. [5]. We show that k -gap planar graphs have unbounded row treewidth and thus cannot be described as a shallow minor of the strong product of a planar graph with a small complete graph, even with $k = 1$. This result is of particular interest since k -gap planar graphs have polynomial expansion [39, Theorem 6.9]. We in fact show a stronger result in terms of local treewidth.

Eppstein [37] introduced the following definition under the guise of the “treewidth-diameter” property. A graph class $\mathcal{G}$ has *bounded local treewidth* if there is a function f such that for every graph $G \in \mathcal{G}$, for every vertex $v \in V(G)$, and for every integer $r \geq 0$, the subgraph of G induced by the vertices at distance at most r from v has treewidth at most $f(r)$. If $f(r)$ is linear, then $\mathcal{G}$ has *linear local treewidth*. For example, every class with bounded layered treewidth (which includes graph classes with bounded row treewidth) has linear local treewidth [30, Lemma 6]. We show that 1-gap planar graphs do not have linear local treewidth, which implies that they have unbounded row treewidth. In fact, we prove an exponential lower bound on the local treewidth.

THEOREM 39. *There exists a constant $c > 0$ such that for infinitely many integers $r \geq 1$, there is a 1-gap planar graph G with radius at most r and $\text{tw}(G) \geq 2^{cr}$.*

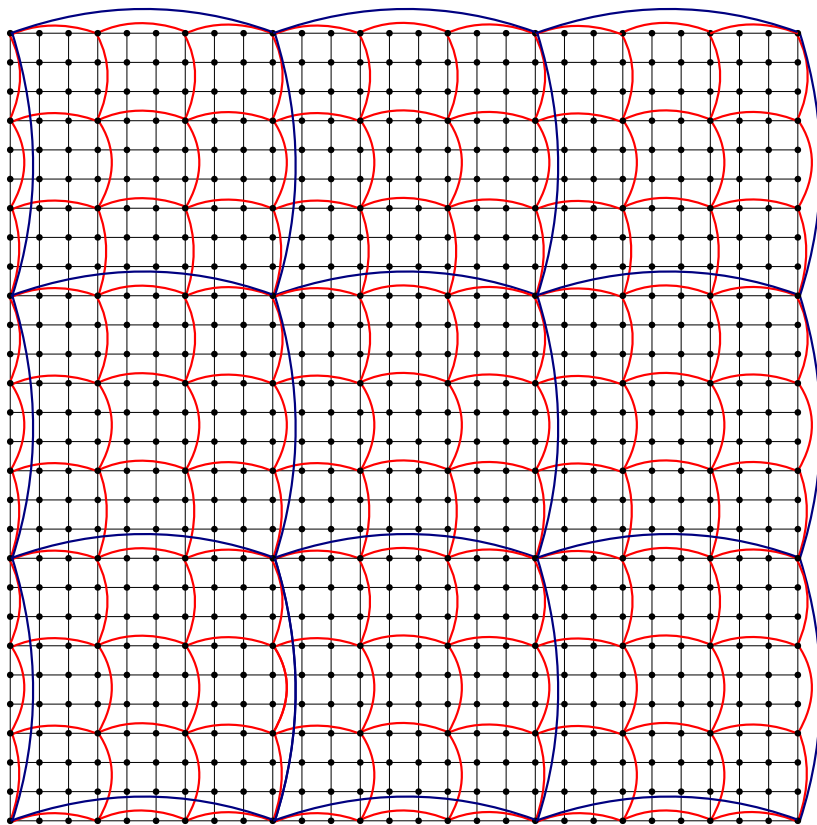
Theorem 39 is implied by the following lemma by setting $n = 2$.

LEMMA 40. *For all integers $n, k \geq 1$, there exists a 1-gap planar graph with treewidth at least $n^{k+1} + 1$ and radius at most $(2k + 1)n + \lceil \frac{k}{2} \rceil + 1$.*

Proof. Let G_∞ be the following infinite graph with $V(G_\infty) := \mathbb{Z}^2$. For $\ell \in [0, k]$, let $A^{(\ell)} := \{(an^\ell, bn^\ell) : a, b \in \mathbb{Z}\}$. Add the *level- ℓ horizontal edges* $(an^\ell, bn^\ell)((a+1)n^\ell, bn^\ell)$ and the *level- ℓ vertical edges* $(an^\ell, bn^\ell)(an^\ell, (b+1)n^\ell)$ to G_∞ for all $a, b \in \mathbb{Z}$.

We claim that G_∞ is k -gap planar (see Figure 10). Embed each vertex $(i, j) \in V(G_\infty)$ at $(i, j) \in \mathbb{R}^2$. Draw the level-0 edges as straight line segments between their end-vertices. For $\ell = 1, 2, \dots, k$ and $a, b \in \mathbb{Z}$, draw the edge $(an^\ell, bn^\ell)((a+1)n^\ell, bn^\ell)$ as a curve between the $y = bn^\ell$ line and the $y = bn^\ell + 1$ line such that it does not cross any horizontal edges or level- ℓ edges. Similarly, draw the edge $(an^\ell, bn^\ell)(an^\ell, (b+1)n^\ell)$ as a curve between the $x = an^\ell$ line and the $x = an^\ell + 1$ line such that it does not cross any vertical edges or level- ℓ edge. This can be done so that for every $j, \ell \in [0, k]$ where $j < \ell$, every level- j horizontal edge is crossed by at most one vertical level- ℓ edge and every level- j vertical edge is crossed by at most one horizontal level- ℓ edge. Charge the crossing to the edge with the lower level. Thus every edge in G_∞ has at most k crossings charged to it and so G_∞ is k -gap planar.

Let $G_n := G_\infty[[0, n^{k+1}] \times [0, n^{k+1}]]$. Then G_n is k -gap planar since it is a subgraph of G_∞ . Moreover, G_n has treewidth at least $n^{k+1} + 1$ since it contains the $(n^{k+1} + 1) \times (n^{k+1} + 1)$ -grid as a subgraph [9]. We now subdivide edges in G_n to obtain a 1-gap

FIG. 10. Construction in the proof of Lemma 40 with $k = 2$ and $n = 3$.

planar graph $\tilde{G}_n$. Replace each edge $uv \in E(G_n)$ that has $c > 1$ crossings charged to it by a (u, v) -path, P_{uv} , on $c + 1$ vertices such that the internal vertices of P_{uv} are between consecutive crossing points that are charged to uv . Let $\tilde{G}_n$ be the 1-gap planar graph obtained from this subdivision procedure. Observe that for every $\ell \in [0, k - 1]$, every horizontal subdivided edge between two vertices in $A^{(\ell)} \cap V(\tilde{G}_n)$ is of the form $(an^{\ell+1}, bn^{\ell})(an^{\ell}, bn^{\ell})$ and every vertical subdivided edge is of the form $(an^{\ell}, bn^{\ell+1})(an^{\ell}, (b+1)n^{\ell})$. Moreover, the level- k horizontal and vertical edges are not subdivided as no crossings are charged to them.

Now $\tilde{G}_n$ has treewidth at least $n^{k+1} + 1$ since it is a subdivision of G_n . So it remains to show that $\tilde{G}_n$ has radius at most $(2k + 1)n + \lceil \frac{k}{2} \rceil + 1$. First, every internal vertex of a subdivided path P_{uv} has distance at most $\lceil \frac{k}{2} \rceil$ to u or v where $u, v \in V(\tilde{G}_n) \cap A^{(0)}$. Let $\ell \in [k]$. Observe that every vertex in $A^{(\ell-1)} \cap V(\tilde{G}_n)$ is of the form $((an + i)n^{\ell-1}, (bn + j)n^{\ell-1})$ where $a, b \geq 0$ are integers and $i, j \in [n]$. Thus $\tilde{G}_n$ contains the following path from $((an + i)n^{\ell-1}, (bn + j)n^{\ell-1}) \in A^{(\ell-1)} \cap V(\tilde{G}_n)$ to $((a+1)n^{\ell}, (b+1)n^{\ell}) \in A^{(\ell)} \cap V(\tilde{G}_n)$ which avoids subdivided paths and has length at most $2n$:

$$((an + i)n^{\ell-1}, (bn + j)n^{\ell-1}), ((an + i + 1)n^{\ell-1}, (bn + j)n^{\ell-1}), \dots, ((a+1)n^{\ell}, (bn + j)n^{\ell-1}), ((a+1)n^{\ell}, (bn + j + 1)n^{\ell-1}), \dots, ((a+1)n^{\ell}, (b+1)n^{\ell}).$$

As such, for each $v \in V(\tilde{G}_n)$, there is a path of length at most $2kn + \lceil \frac{k}{2} \rceil$ from v to some vertex in $A^{(k)} \cap V(\tilde{G}_n)$. Finally, the induced subgraph $\tilde{G}_n[A^{(k)}]$ has radius at

most $n+1$ since it is isomorphic to the $(n+1) \times (n+1)$ grid. Therefore $\tilde{G}_n$ has radius at most $(2k+1)n + \lceil \frac{k}{2} \rceil + 1$. $\square$

We conclude this subsection with the following open problem: do k -gap planar graphs have bounded local treewidth?

5.2. RAC, fan-crossing free, and k -quasi planar graphs. Recall that an embedded graph G is:

- k -quasi planar if every set of k edges do not mutually cross;
- fan-crossing free if for each edge $e \in E(G)$, the edges that cross e form a matching; or
- RAC if each edge is drawn as a straight line segment and all crossings are at right angles.

A graph is k -quasi planar, fan-crossing free, or RAC if the graph is respectively isomorphic to an embedded graph that is k -quasi planar, fan-crossing free, or RAC. Recall that a graph class $\mathcal{G}$ is somewhere dense if there exists an integer $r \geq 0$ such that every graph H is an r -shallow minor of some graph $G \in \mathcal{G}$ (see [67]). Drawing on known results, Brandenburg [15] proved that for every graph G , the 3-subdivision of G is a RAC graph; the 2-subdivision of G is fan-crossing free; and the 1-subdivision of G is 3-quasi planar. This implies that each of these classes is somewhere dense. Note that Eppstein [38] previously observed that the class of RAC graphs is somewhere dense. Since graph classes with bounded row treewidth have polynomial expansion [34], these beyond-planar classes have unbounded row treewidth. Thus, they cannot be described as a shallow minor of the strong product of a planar graph with a small complete graph. Additionally, Brandenburg [15] asked what is the queue-number and stack-number of these classes. Since graph classes with bounded stack-number or bounded queue-number have bounded expansion [69], we conclude that RAC, fan-crossing free, and k -quasi planar graphs (where $k \geq 3$) have unbounded stack-number and unbounded queue-number.

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